

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics

In partial fulfillment for the requirements in Capstone 1 and
Research Subject

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CHAPTER 1

1.1 Background of the Study

In today's fast-paced world, businesses across all sectors are shifting to digital platforms to keep up with the growing expectations of modern consumers. The food service and baking industry in the Philippines is experiencing this exact transition, as tech-savvy individuals increasingly rely on mobile devices and computers to browse menus, place orders, and settle transactions remotely. Because of this shift in consumer behavior, small local micro-enterprises that continue to rely on manual, paper-based processes face severe operational challenges. Traditional, non-automated methods often lead to critical inaccuracies, limited market reach, and an inability to meet daily customer expectations. For small community bakeries, operating without a modern web presence means losing potential sales to larger competitors who offer instant online ordering, clear pricing, and transparent communication.

The core issue stems from how order information is captured, stored, and processed during daily operations. In a digital commerce platform, data entries are structured, validated, and processed automatically by backend logic.

Conversely, traditional store setups rely heavily on loose paper slips, handwritten notebooks, and human memory. This lack of automated validation leaves daily transactions highly vulnerable to misplaced records, delayed order tracking, and operational confusion. When a small business operates entirely without digital validation, everyday tasks like taking an order, calculating a price, or scheduling a delivery turn into stressful, repetitive manual chores that consume huge amounts of time and lower staff productivity. For community-scale enterprises, continuing to operate within a purely analog framework creates a technical barrier that cuts them off from the modern digital marketplace, making it increasingly difficult to sustain consistent growth or build long-term customer loyalty.

C&N Bakery is a local establishment serving a growing customer base within a competitive community market. The bakery offers a wide selection of daily baked goods, including fresh breads, pastries, cakes, and traditional local delicacies . In addition to selling standard retail items over the counter, the bakery specializes in custom cake orders made for special events like birthdays, weddings, debuts, and anniversaries. These custom orders represent a significant portion of the business's overall revenue and require careful planning and precise execution.

Despite managing a diverse product menu and handling many daily customers, C&N Bakery handles its entire workflow through manual mechanisms, such as walk-in counter visits, phone calls, and disorganized social media messages on Facebook. The business operates without a centralized digital system to organize its daily activities, leaving crucial customer choices and transaction records disorganized across different notebooks, paper slips, and informal message threads.

When a business operates without a single central system to store its records, information easily gets lost or fragmented. At C&N Bakery, a single custom transaction often leaves an unstructured data trail across multiple communication channels. Design layouts, delivery deadlines, special instructions, and payment receipts are frequently recorded in completely different places. Because these details are isolated from one another, staff members must manually piece together the customer's requirements before baking can start. This manual approach consumes massive amounts of time and makes day-to-day operations chaotic, turning simple routine tasks into exhausting manual procedures that drain the small business's daily energy and workforce productivity.

Managing custom cake orders is inherently far more complex than selling pre-baked goods over the counter. A single custom cake

request requires capturing and tracking multiple precise attributes simultaneously, including cake dimensions, tier structures, flavor variations, frosting types, color schemes, personalized dedication messages, and visual reference photos. At the same time, the bakery must carefully coordinate pickup or delivery dates to guarantee product freshness and structural integrity during transit. Because C&N Bakery currently relies on handwritten logs, loose paper receipts, and unorganized chat histories, staff members can easily misread or forget specific design preferences by the time production begins in the kitchen. Furthermore, writing down fulfillment dates manually in physical logbooks frequently causes human error, resulting in accidental double-bookings that overload the kitchen team beyond their daily shift capacity.

This operational vulnerability is deeply tied to the problem of manual transcription. When a customer outlines their custom selections through casual communication channels, that information exists as an unformatted text string. As that request moves from the front counter staff to the production area, the information is manually rewritten onto paper order tags or kitchen whiteboards. Every time a human manually copies, interprets, or rewrites an order detail, the risk of human error occurring increases significantly. Essential design choices can

easily be misread, miscalculated, or completely forgotten during the transfer. This manual workflow creates a weak operational loop where the original intent of the customer rarely matches the actual physical instructions given to the baking staff on the kitchen floor.

Every time an order detail is manually transcribed, the risk of miscalculation increases. Essential design preferences can easily be misinterpreted, miscalculated, or left out entirely. When a cake is prepared with incorrect specifications, it triggers a chain reaction of operational failures across the business, resulting in wasted raw ingredients, disrupted baking schedules, lost revenue, and damaged customer trust. The financial impact of these errors is particularly severe due to the rigid nature of the baking process itself. Unlike quick-service meals that can be replaced in minutes, custom cakes require hours of uninterrupted baking, cooling, and detailed decorating. If an error is noticed late in the process, the entire cake often has to be discarded, delaying every other order waiting in the kitchen queue and creating a backlog that ruins the entire work shift.

These internal difficulties are further magnified by the rapid adoption of digital payment solutions in the Philippines, particularly electronic wallets like GCash and online bank

transfers. Modern consumers expect clear pricing, immediate feedback, and simple remote payment options. At C&N Bakery, operating manually creates major communication bottlenecks. When a customer sends a message asking for a price quote on a custom cake design, kitchen staff must pause their physical work to manually calculate costs line-by-line based on cake size, tiers, and add-ons. This manual step leads to long response delays, leaving potential buyers waiting in digital queues for simple information. In an era where buyers expect instant answers, waiting hours for a message reply causes many customers to abandon their inquiries and purchase from larger, automated competitors instead.

Furthermore, once an order is placed, customers currently lack a clear way to monitor their order status in real time. Because there is no online order tracker, customers are forced to call or message the store repeatedly to ask if their cake is being prepared, baked, or ready for pickup. This constant back-and-forth communication gap creates a major operational barrier inside the store. Front counter staff and kitchen personnel do not share a synchronized information system, making it hard to track order priorities and preparation sequences. Staff members must constantly stop their physical duties to verbally verify order statuses with the bakers. This constant

workplace interruption wastes valuable labor time, stresses out the staff, and leads to scheduling mistakes that leave waiting customers frustrated.

Operating without a secure web system also introduces serious data security risks and administrative obstacles. On the customer side, users lack a secure account interface to review their past orders or monitor active deliveries. For administrative personnel, operating without a secure, centralized administrative dashboard means keeping financial records, customer contact details, and transaction histories in paper notebooks stored directly inside the active kitchen. These paper logs are constantly exposed to water spills, flour dust, grease stains, heat degradation, and physical loss. If a single notebook is misplaced, damaged, or destroyed, the bakery permanently loses its operational records with no way to recover the data. Storing customer addresses and phone numbers on loose paper slips also presents privacy concerns, while management is left without historical records to evaluate monthly revenue, track sales growth, or identify best-selling product trends.

To resolve these operational vulnerabilities, this study will develop the Web-Based Order, Tracking, Customization, and Sales Analytics System for C&N Bakery. The proposed web application will establish a centralized digital platform engineered to

streamline order taking, dynamically balance daily kitchen capacity limits, track production stages across a clear lifecycle, and compile historical records into clear visual sales reports. The system will provide dedicated user accounts for registered customers while maintaining open browsing access for guest users.

Through this proposed platform, customers will be able to customize their cake selections, view automated cost calculations, submit reference references, and select available fulfillment dates backed by real-time schedule checks to prevent kitchen overbooking. Administrative staff will be provided with a centralized management dashboard to verify order payments, update production progress in real time, and view visual sales analytics. Built using cost-effective cloud database technology, this practical architecture will keep operational expenses manageable, safeguard store transaction records, eliminate manual data entry errors, optimize kitchen productivity, and provide C&N Bakery with an efficient, reliable web presence tailored to its daily business needs.

1.2 Statement of the Problem

1.2.1 General Problem Statement

The main problem facing C&N Bakery was its heavy reliance on manual paper logs, physical wall calendars, and loose social media chat threads to run daily business activities. Because the bakery lacked a central web platform, store data remained unstructured across notebooks, loose paper slips, and chat apps.

This manual process led to delayed replies to customer inquiries, frequent cake design mistakes, kitchen overbooking, and a lack of reliable sales records. Storing customer contact details and payment records in unsecured paper notebooks also exposed the business to privacy risks and lost data. To resolve these operational issues, the proposed Web-Based Order, Tracking, Customization, and Sales Analytics System for C&N Bakery will create a central online platform to automate order taking, coordinate kitchen schedules, track baking progress, and generate real-time sales reports.

1.2.2 Specific Problem Statements

To address the main problem, the specific operational issues at C&N Bakery were divided into three main areas:

1. Ordering and Customer Communication Problems

- **Communication Delays:** Customers waited hours for replies about menu items, cake availability, or price estimates because front-counter staff answered online messages while serving walk-in store lines.
- **Inconsistent Custom Pricing:** Custom cake prices were calculated manually without standard formulas, leading to inconsistent price quotes, calculation errors, and a lack of clear price breakdowns for buyers.
- **Design Misunderstandings & Copying Errors:** Staff manually copied custom cake details from casual chat messages onto paper kitchen slips. This manual step frequently led to misread colors, wrong tier sizes, or missing cake dedications, causing customer rejections and wasted ingredients.
- **Lack of Easy-to-Find Guidelines:** The bakery had no online page showing store rules, advance ordering deadlines, or payment steps, which forced staff to answer the same simple questions repeatedly and made buyers hesitant to place orders.

2. Kitchen Scheduling and Production Issues

- **Kitchen Overbooking:** Completion dates were tracked on a single paper wall calendar. Without an automatic schedule check, staff routinely accepted new orders past daily kitchen limits, leading to rushed work, lower cake quality, and late pickups.
- **Disorganized Production Tracking:** Kitchen staff lacked a central visual board to track order stages. Staff constantly interrupted bakers to ask for order status updates, which slowed down baking and caused workplace confusion.
- **Pickup & Delivery Scheduling Mix-ups:** Relying on handwritten pickup slips led to misread fulfillment dates, causing cakes to be made too early (losing freshness) or forgotten completely until the customer arrived.
- **Lost Order Records at Hand-off:** Completed orders were logged manually in counter notebooks. Missing or unfiled paper slips created immediate gaps in daily sales records, making daily cash counts inaccurate.

3. Data Security, Privacy, and Bookkeeping Risks

- **Unsecured Customer & Financial Records:** Customer phone numbers, addresses, and sales notebooks were kept in

physical logs exposed to kitchen spills, damage, or unauthorized access, with no digital backup.

- **Lack of Business Insights:** The bakery owner spent hours manually adding up paper receipts at the end of every month, leaving management without clear figures on net profits, popular products, or seasonal sales trends.

1.3 Objectives of the Study

This project aims to build a complete web-based platform for C&N Bakery that handles online ordering, custom cake designs, order tracking and sales analytics. The system will move C&N Bakery away from manual paper processes by bringing their ordering, custom design, and scheduling workflows into one easy-to-use site, featuring separate accounts for customers and owner/staff to keep operations secure and organized.

Core Objectives of the System

- Make browsing and ordering bakery products online simple by offering a filterable menu, a shopping cart, and secure customer accounts.
- Provide an interactive custom cake builder that lets buyers choose layers, sizes, flavors, frostings, and designs with automatic real-time pricing.

- Allow buyers to upload reference photos and dedication messages during checkout to ensure exact design details.
- Give customers a live step-by-step progress bar (Preparing, Baking, Packing, Ready) to track active orders, view order history, leave reviews, or manage pending requests.
- Provide bakery staff with a central dashboard to review incoming orders, check payment screenshots from Facebook Messenger, and approve or decline requests.
- Send automatic email notifications to customers whenever their order status is updated by the bakery team.
- Give bakery management an operations calendar to set daily order limits, block full dates, and toggle pickup or delivery availability.
- Give staff full control over the online catalog to add, edit, or remove menu items, tweak cake options, and highlight featured products.
- Generate visual sales analytics charts showing monthly revenue trends, top-selling items, and product category splits over custom date ranges.

1.4 Scope and Delimitation of the Study

This study focuses on designing, building, and testing a web-based ordering and management platform for C&N Bakery to

shift its daily store operations into a central digital system. The project aims to reduce reliance on paper notebooks, kitchen calendars, and scattered social media messages by providing a single web application divided into a public storefront for customers and a secure backend portal for store staff.

1.4.1 System Setup and Accessibility

The platform will be built as a mobile-friendly web application, meaning users and staff will not need to download or install any apps from an app store. The screen layout will automatically adjust to fit different screen sizes, including desktop computers, laptops, tablets, and smartphones. The website will work across standard web browsers such as Google Chrome, Mozilla Firefox, and Apple Safari, and will require an active internet connection to communicate with a cloud database in real time.

1.4.2 User Roles and Access Controls

The study considers two primary user groups to handle day-to-day interactions:

- **Bakery Customers:** Unregistered guests can browse the menu, read product descriptions, check FAQs, and view feedback left by past customers. Registered users gain access to a

personal account area where they can build custom cake requests, submit orders, attach design pictures, and track their active orders.

- **Administrative and Kitchen Staff:** The store owner and kitchen crew share a single backend link where everyone logs in to manage daily shop operations together. Through this shared workspace, staff can view order specifications, monitor kitchen schedules, update baking progress, manage product prices and availability, adjust daily order caps, and view automated sales summaries.

1.4.3 Scope of the Customer Portal

The customer-facing portion of the platform focuses on providing an accessible online shopping experience:

- Customers will be able to browse the menu using category filters to quickly find cakes and other products.
- Customers will be able to design custom cakes by picking their preferred sizes, tiers, flavors, and frostings while seeing real-time price estimates.
- Customers will be able to upload reference photos and type in dedication messages so the kitchen knows exactly what to bake.

- Customers will be prevented from selecting fully booked dates at checkout, as the calendar automatically blocks days that reach maximum daily capacity.
- Customers will be able to manage their shopping cart easily by adjusting quantities or removing items before heading to checkout.
- Customers who are logged in will be able to track their cake's current status on the site without needing to call or message the store.
- Customers will receive automatic email updates whenever their order status changes.
- Customers will be able to view their order history once orders are completed or cancelled, and leave ratings or reviews
- Customers will be able to update their account details, such as changing their password or email address, whenever they need to

1.4.4 Scope of the Administrative and Kitchen Dashboard

- The main dashboard will display revenue summaries, pending requests, and current kitchen tasks right when staff log in.

- Staff will be able to search, filter, and sort incoming orders by customer name, pickup date, or payment status.
- An order view that displays all customer selections—including reference photos, flavor choices, size dimensions, and dedication text—on one screen to avoid kitchen mistakes.
- A status manager that lets kitchen staff update baking progress stage-by-stage and verify customer payment screenshots.
- A switch that allows staff to hide out-of-stock items from the public storefront during ingredient shortages.
- Staff/Owner will be able to review all customer selections—including reference photos, flavor choices, size dimensions, tier configurations, and dedication text—on a single order screen to prevent kitchen mistakes.
- Staff/Owner will be able to manage the product catalog by adding, editing, or removing menu items, categories, pricing, and cake builder options.
- A calendar management tool that lets the owner and staff set daily order limits and toggle delivery or pickup options on or off for specific dates to prevent kitchen overload.

- The system will automatically convert sales records into visual charts to help the owner track revenue trends and identify top-selling products, while also allowing sales data to be exported as CSV files for easy record-keeping.

1.4.5 Delimitation of the Study

To keep the project focused and manageable for a small local bakery, the following capabilities are explicitly excluded from this study:

- **No Native Mobile Apps:** The project is strictly a web application and will not involve publishing apps on Google Play or the Apple App Store.
- **No Automated Payment APIs:** The system will not integrate with direct banking APIs or automated payment gateways. Staff will verify GCash or bank transfers manually by reviewing uploaded payment screenshots.
- **No Third-Party Courier Integration:** The system will not connect to external delivery APIs (like Grab or Lalamove). Arranging delivery drivers remains a manual task handled outside the platform.
- **No Raw Ingredient Inventory Tracking:** The system will not track remaining weights of raw baking materials (such as

flour, sugar, or eggs). Stock management will rely on staff checks and manual menu visibility toggles.

- **No Payroll or HR Modules:** The platform focuses solely on orders, kitchen schedules, and sales reporting; it will not track employee hours or calculate staff salaries.
- **No Supplier Management:** The platform will not track ingredient suppliers or wholesale purchase orders.
- **No Offline Mode:** The system relies entirely on a cloud database and will require an active internet connection to function.
- It will not refund or order cancellations once a staff member or admin officially confirms the order.

1.5 Significance of the Study

Developing the Web-Based Order, Tracking, Customization, and Sales Analytics System for C&N Bakery will bring practical value to everyone involved in the daily operations and growth of the business. As local food businesses move online, neighborhood bakeshops need modern tools to stay competitive. This study shows how a local bakery can smoothly transition away from manual paperwork. By replacing easily lost paper logbooks, fading sticky notes, and messy social media chat threads with a single web platform, this project creates an organized ordering process. It ensures clear pricing for buyers, protects kitchen

staff from order overbooking, and keeps business records safe from physical loss.

By modernizing these daily tasks, this study provides clear, long-term benefits to the following stakeholders:

1. Bakery Customers and Guests

Traditionally, buyers ordering custom cakes or local delicacies deal with slow chat replies, missing design details, and unclear pricing. The system directly helps customers by providing:

- **Clear and Upfront Pricing:** Customers see instant price updates as they choose cake sizes, flavors, and designs, removing price confusion.
- **Accurate Design Submission:** Buyers can upload sample photos and type exact custom text notes, making sure the final cake matches what they asked for.
- **Self-Service Order Tracking:** Customers can log into their accounts anytime to check their order progress (such as preparing, baking, or ready for pickup) without waiting for a reply.
- **Easy Menu Browsing:** The online catalog lets customers browse products and plan orders conveniently from any device with internet access.

2. Production and Kitchen Staff (Decorators and Bakers)

Kitchen staff working with paper logs face daily pressure to track tight deadlines and complex cake designs. This project organizes the kitchen workflow by providing:

- **Workload Protection:** The system limits how many orders can be booked per day, preventing the kitchen team from taking on more work than they can handle.
- **Order Accuracy:** Decorators can view complete cake details and reference photos on a single screen, avoiding mistakes caused by searching through loose paper slips or long chat messages.
- **Step-by-Step Progress Tracking:** Bakers can update order progress step-by-step, moving each cake clearly from preparation to baking, packing, and pickup.
- **Quick Menu Updates:** Staff can instantly hide out-of-stock items on the website whenever raw ingredients run short, preventing customers from ordering items the kitchen cannot make.

3. Executive Management (The Bakery Owner)

Running a custom bakery without a digital system makes record-keeping and financial planning difficult. This platform helps the owner manage daily operations by offering:

- **Automated Sales Records:** The system organizes customer payments, down-payments, and order receipts into a digital record, reducing math errors.
- **Visual Sales Trends:** The owner can view visual sales charts to see monthly earnings and top-selling products, making it easier to plan ingredient purchases and budget ahead.
- **Data Privacy and Protection:** Password logins keep business sales numbers safe from unauthorized access.
- **Better Business Decisions:** Clear sales summaries help the owner make smart choices about menu items, prices, and holiday promotions.

4. C&N Bakery as a Business

Moving to a digital platform builds a professional brand image and protects the bakery's reputation. The store benefits through:

- **Less Ingredient Waste:** Clear order records and confirmed payment receipts ensure the kitchen never wastes costly baking ingredients on unconfirmed or misread orders.
- **Stronger Local Brand:** Modern digital conveniences give this local bakery an edge over nearby competitors.

- **Room to Grow:** Automated order taking and tracking allow the shop to handle higher order volumes during holiday seasons without confusing staff or missing deadlines.

5. Future Systems Developers

Standard online shopping carts do not handle custom cake options or daily calendar limits well. This study helps future software developers by providing:

- **Custom Order Form Example:** A practical reference for building custom forms that adjust prices in real time based on user choices.
- **Capacity Limit Logic:** A template for setting up automated calendar limits that block pickup dates when daily kitchen capacity is reached.

6. Future Academic Researchers

This study offers clear documentation on how a small retail business moves from paper-based operations to a Web-based system. It serves as a real-world reference for researchers studying small business automation and provides a foundation for future student projects (such as adding SMS notifications or delivery app integrations).

7. The Local Food Service Industry

This project demonstrates that simple, effective web tools are accessible to small, local food vendors without requiring large software budgets. Showing how digital tools improve reliability encourages other neighborhood shops to modernize as well.

8. BSIT Students

For Information Technology students, this manuscript provides a practical guide for breaking down manual business problems into working software features. It shows how to turn real-world store operations into database structures, user interfaces, and role-based security rules.

1.6 Definition of Terms

To ensure a clear and accurate understanding of the technical concepts, system operations, and key components discussed throughout this study, the following terms are defined both how they operate in standard practice and (how they will be applied inside the C&N Bakery platform):

1. Web-Based System

- A software application that runs on a central server and is accessed through a standard web browser over an internet

connection without needing to download or install local software.

- This refers to the online ordering and management software built for C&N Bakery. It will allow customers, kitchen staff, and the store owner to browse products, place custom cake orders, track active progress, and view sales reports using any web browser on a phone, tablet, or computer.

2. User Authentication and Account Access Control

- A security process that verifies a user's login credentials and checks their account permissions before granting access to protected system areas.
- This will serve as the system's security checkpoint during login. Based on the user's assigned role, the system will automatically route customers to the public storefront, kitchen staff to production schedules, and the bakery owner to secure financial records.

3. Interactive Custom Cake Builder

- An interactive web page that lets users choose, customize, and modify specific options for a product before placing an order.
- This will be the cake design page on the website where customers create custom orders. Buyers will select options

such as tier counts, sizes, cake flavors, and frostings, while typing custom dedication text and uploading reference photos.

4. Automated Pricing Calculator

- A background script that immediately updates total prices whenever a user changes a selection on screen.
- This will be the calculation script operating inside the Custom Cake Builder page. The moment a customer chooses an option—such as a larger cake size or a special frosting—the script will update the final price quote in real time, removing the need for manual price quotes over chat or phone calls.

5. Online Menu Catalog

- An organized digital display that shows a business's product list in clear, searchable categories for easy browsing.
- This will serve as the digital menu for C&N Bakery's product lineup. It will automatically categorize items—such as customized cakes, cupcakes, *kakanin*, and *sapin-sapin*—and will feature a search bar alongside price filters to help shoppers find treats within their budget.

6. Private Customer Dashboard and Progress Tracker

- A private account section that displays user details, past transaction history, and live status updates on active requests.
- This will be the personal account page for registered C&N Bakery customers. Users will be able to review past orders and track active custom requests through a visual status bar that updates as kitchen staff order progress.

7. Sales Analytics Charts

- A reporting tool that converts raw stored database records into visual graphs to show business performance trends over time.
- This reporting tool will gather completed sales records and display them in visual charts: a Monthly Revenue Graph to track overall sales growth, and a Top-Selling Products Graph to show which items bring in the most revenue.

8. Custom Cake Option Status Control

- An administrative setting that allows system managers to toggle the availability status of individual customization choices on the platform.
- *This is basically an on-and-off switch for cake options in our admin panel. If the kitchen runs out of a certain flavor or layer, staff can just set it to 'Disabled' so customers can't pick it up on the website while it's out of stock.*

9. Auto-Lock Booking Calendar

- An automated scheduling rule that counts daily reservations against a preset limit and automatically closes calendar dates once the maximum limit is reached.
- This will function as the system's workload gatekeeper. It will continuously count incoming custom cake orders per date and automatically lock that day on the customer booking calendar as soon as it hits the bakery's daily limit, protecting kitchen staff from accidental overbooking.

CHAPTER 2

2.1 Foreign Studies

Mobile e-commerce platforms and cloud backends play a central role in simplifying everyday tasks for small, independent food businesses. Liang et al. (2021), in *A Mobile E-Commerce System for Local Bakery Management using Flutter and Firebase* (IEEE Xplore, ISSN: 2169-3536), examined how mobile software and cloud database tools help local bakeries manage their daily operations. The authors observed that small bakeries frequently run into trouble when they rely on manual paper logs, paper slips, or unorganized social media chat threads to write down incoming customer orders. When sales volume increases during busy weekends or seasonal holidays, paper order slips easily get misplaced, misread, or damaged near kitchen prep tables, which invariably leads to forgotten cake orders, wrong delivery times, and confused baking staff.

To solve this persistent operational problem, the researchers built a cross-platform application that allowed customers to browse bakery catalogs, select items, and place orders through their smartphones or web browsers. They connected the frontend user application directly to a real-time cloud database system. Whenever a customer submitted an order online, the web

application instantly updated the store manager's control panel without requiring anyone to manually refresh the browser page. The evaluation revealed that real-time database syncing successfully eliminated duplicate order records, cut down order processing delays, and kept store staff instantly informed about newly incoming purchases.

This study is highly relevant to the proposed platform because it proves that real-time database tools keep customer orders and kitchen preparation schedules synchronized without human transcription errors. Traditional manual logs create unnecessary delays and risks for local bakeshops, whereas a synchronized cloud database ensures that custom order details are preserved accurately. The proposed project adapts a similar real-time database architecture to ensure that whenever a customer places an order or customizes a cake online, the bakery manager's administrative dashboard updates immediately, preventing lost orders and missed pickup deadlines.

Interactive customization tools are also essential when a food business sells made-to-order goods like specialized event cakes. Rajaput et al. (2022), in *Bakery Shop Cake Customization Management System* (Genesis Global Publication, ISSN: 2582-7421), investigated how interactive ordering systems reduce miscommunication between cake buyers and bakers. The authors

pointed out that ordering a custom cake is far more complicated than buying pre-baked, standard items off a retail display shelf. Customers usually want specific cake dimensions, multi-tier structures, unique flavor combinations, custom frosting colors, specific decorative toppings, and short text inscriptions written on top of the cake. When these intricate details are discussed over simple phone calls or open-ended messaging apps, misunderstandings happen frequently, leading to disappointed customers and wasted baking ingredients.

To address this issue, the researchers designed an interactive web builder that guides customers through a step-by-step customization process. Instead of typing long, unorganized text messages, buyers select options from organized visual menus, such as choosing the number of cake tiers, selecting base sponge flavors, picking frosting styles, and uploading reference images for the baker to inspect. The testing results confirmed that offering structured choices through a dedicated web interface significantly improved overall order accuracy, saved administrative time for store staff, and reduced customer complaints regarding incorrect cake designs.

This foreign study provides a practical visual blueprint for the proposed platform's custom cake module. By offering clear drop-down choices, selection cards, and upload input fields

directly on the website, the proposed platform will help customers specify their exact cake design upfront without needing back-and-forth messaging. This structured approach eliminates guesswork for the local baker, ensuring that the final custom cake matches the buyer's exact design specifications while saving time for both parties during the ordering process.

Web system architecture must also accommodate the clear operational differences between ready-made baked goods and custom orders. Mali (2020), in *Fast and Efficient Cake Shop Management System Using Web-Based Architecture* (Technoscience Academy, ISSN: 2395-6011), studied how structured web applications streamline daily management tasks for small cake shops. The author explained that standard bakery inventory, such as pre-baked breads or display-case pastries, follows a simple sales process where stock counts are subtracted immediately upon purchase. On the other hand, custom cakes require dedicated labor planning, raw material preparation scheduling, and multi-stage order status tracking over several days.

When a store manager tries to track both standard off-the-shelf sales and custom order schedules in the exact same physical logbook, daily bookkeeping becomes messy, time-consuming, and confusing. The researcher built a web architecture that

separated ready-made retail sales from custom order workflows into two distinct administrative dashboard views. The findings confirmed that separating these two product pipelines reduced administrative stress for business owners, simplified daily transaction records, and improved overall operational response times during peak business hours.

This study offers useful architectural guidance for the backend layout of the proposed system. Custom cakes demand a completely different preparation and tracking workflow compared to everyday baked goods, making separate management views necessary for efficient store operations. Mali's research justifies dividing the proposed platform's administrative panel into separate modules for regular daily menu orders and custom cake preparations, allowing the local bakery owner to monitor immediate retail sales and upcoming custom baking schedules simultaneously without operational confusion.

Managing kitchen labor limits requires automated order scheduling to protect bakery staff from operational overload and burnout. Sharma et al. (2020), in *Smart Scheduling and Dynamic Capacity Allocation in Custom Order E-Commerce Systems* (Foundation of Computer Science, ISSN: 0975-8887), analyzed automated scheduling logic and capacity management for businesses that make hand-crafted or custom-built products. The

authors explained that custom manufacturing relies heavily on human labor hours, which means every kitchen or workshop has a strict maximum daily output limit. In custom cake baking, taking on more detailed orders than the staff can physically decorate in a single day leads to rushed workmanship, late deliveries, compromised quality, and burned-out decorators.

To solve this widespread overbooking issue, the researchers developed a dynamic scheduling feature that calculated daily production limits based on available labor hours and order complexity. When total accepted orders reached the store's maximum threshold for a specific date, the web system automatically locked that calendar date on the online checkout page, preventing new customers from selecting or booking that day. This automated process protected store owners from accidentally taking on more work than their kitchen crew could physically handle during peak event seasons.

This study provides the primary theoretical and structural backing for the Smart Scheduling feature of the proposed platform. Custom cake baking is a labor-intensive process, and local bakeries must enforce strict daily limits to maintain quality and meet delivery deadlines. By implementing automatic calendar slot limits based on the concepts established by Sharma et al., the proposed system will protect local bakery staff from

overbooking during busy dates and major holidays, ensuring that every accepted order is delivered on time with high quality.

Visual sales dashboards help small business owners track earnings without having to read through long, complicated numerical spreadsheets. Kumar et al. (2021), in *Real-Time Sales Analytics and Visualization Frameworks for Small-Scale Retail Businesses* (Elsevier Science, ISSN: 0164-1212), examined how automated data visualization tools help small enterprise owners make practical business decisions. The authors observed that small food enterprise owners rarely have the time or specialized analytical skills to review long line-item sales logs or extract numbers from manual accounting ledgers. As a result, many small shop managers make raw material purchases and menu pricing decisions based on rough guesswork rather than real financial data.

To solve this decision-making hurdle, the researchers developed an analytics framework that automatically collects raw database records and converts them into clean visual charts, such as monthly revenue bar graphs, daily sales volume trends, and pie charts showing popular product choices. The evaluation revealed that visual graphic displays allowed non-technical store owners to quickly spot top-selling menu items, recognize seasonal

revenue surges, and plan raw material purchases well ahead of time.

This research directly informs the design and layout of the Sales Analytics module in the current project. Small bakery managers need clear, at-a-glance financial insights to manage raw ingredient costs and identify popular cake choices without spending hours on bookkeeping. By studying the visual chart layouts proposed by Kumar et al., the proposed system will provide local bakery managers with an intuitive, easy-to-read dashboard that automatically translates historical order data into clear visual reports, helping them make smarter business and purchasing decisions.

2.2Local Studies

Local information system development plans provide valuable insight into how Philippine bakeshops transition from traditional paper workflows to digital platforms. An information system plan for Manolette Bakeshop, documented in the Neliti Philippine Research Repository (Neliti Repository, ID: 492104), evaluated the operational challenges faced by established community bakeries in the Philippines. The study highlighted that many local bakeshops continue to rely on physical notebooks, paper receipts, and verbal communication between counter staff and kitchen workers. When handling custom cake requests, staff members had to write down specific design choices by hand, which frequently resulted in misplaced order slips, forgotten customer notes, and overall operational delays.

To solve these persistent administrative issues, the study outlined a structured digitalization roadmap focused on centralizing order tracking, digitizing customer records, and establishing clear operational workflows. The roadmap provided a step-by-step model for replacing manual order logs with an interconnected digital system that connects counter staff, bakery owners, and kitchen decorators. The results of the plan showed that adopting digitized management tools reduced human

record-keeping errors, eliminated lost receipts, and streamlined daily communication across departments.

This local study is directly relevant to the proposed platform because it proves that traditional Philippine bakeries need structured digital management tools to replace error-prone paper workflows. Operating a custom cake business through manual notebooks leaves room for expensive mistakes and customer dissatisfaction. The proposed project adapts the core principles of this digitalization roadmap by providing a centralized web platform that stores every custom cake request in a digital cloud database, ensuring that custom order details are preserved accurately without physical paperwork.

Regional studies further emphasize the importance of online ordering and inventory integration for provincial bakeries. Santos et al. (2021), in *Development of an Online Ordering and Inventory System for Local Bakeshops in Laguna* (Philippine e-Journals, ISSN: 2094-1749), examined how web-based software helps small-scale food enterprises across Laguna modernize their sales process. The authors noted that provincial bakeshops struggle to balance raw material inventory with unpredictable daily walk-in sales and custom event orders. When ingredient tracking is handled manually, bakeries often run out of essential baking supplies—such as specialized cake flour, food

colorings, or fondant—right before completing major custom orders.

To address these inventory bottlenecks, the researchers designed an integrated web application that connected an online ordering portal directly to an automated stock deduction panel. Every time a customer placed an order online, the system automatically checked available ingredient stock and updated inventory counts in real time. The study confirmed that combining online ordering with automated inventory tracking minimized stock shortages, reduced ingredient waste, and improved overall fulfillment speed for local bakery owners.

This research provides functional guidance for the proposed system by demonstrating how online ordering platforms must seamlessly sync with kitchen availability to keep operations running smoothly. Unplanned ingredient shortages ruin baking schedules and delay customer deliveries. By understanding the inventory challenges identified by Santos et al., the proposed platform will incorporate real-time order processing that helps local bakery managers monitor their daily production capacity and order volumes effectively, ensuring timely order fulfillment.

Web-based customer relationship management tools tailored for local micro-enterprises also play a crucial role in improving

customer retention. Garcia et al. (2022), in *Web-Based Custom Cake Ordering and Customer Relationship Management System for Micro-Enterprises in Cavite* (Journal of Philippine Information Technology Research, ISSN: 2782-8883), investigated how specialized ordering software benefits home-based and micro-bakery businesses in Cavite. The authors pointed out that micro-bakery owners spend an excessive amount of time answering repetitive inquiries on social media apps regarding cake flavors, sizes, pricing estimates, and available pickup dates. This constant back-and-forth messaging takes time away from actual cake baking and decorating.

To solve this issue, the researchers developed a dedicated web platform equipped with an interactive custom cake builder and an integrated customer database. Customers could select their desired cake specifications online, calculate estimated costs automatically based on selected features, and track their order status without needing to text the owner directly. The study revealed that providing a self-service custom ordering portal reduced administrative workloads for micro-enterprise owners while significantly improving customer satisfaction rates.

This study serves as a strong functional model for the proposed project, reinforcing the value of providing a self-service online cake configurator that frees up the baker's time. Relying

solely on social media chat threads causes severe operational bottlenecks for local Philippine cake decorators. By adopting the interactive ordering framework from Garcia et al., the proposed platform will allow local customers to select their preferred cake options directly on the website, reducing repetitive chat inquiries and allowing the baker to focus on cake preparation.

Workflow management and order status tracking are equally critical during the kitchen production phase. Aquino et al. (2020), in *Order Tracking and Workflow Management Platform for Community-Based Bakeries in Central Luzon* (Philippine Journal of Science and Technology, ISSN: 0115-2130), analyzed how structured order status tracking improves operational productivity in community bakeries across Central Luzon. The researchers observed that once a custom order is accepted, kitchen staff often lose track of which cakes need immediate preparation, leading to late pickups, rushed decorating, and customer dissatisfaction.

To resolve these production delays, the authors built a central workflow tracking system that categorized each order into distinct status stages, such as *Pending*, *In Progress*, *Decorating*, and *Ready for Pickup*. Kitchen staff updated the order status in real time through an admin tablet view, allowing

customers to view the exact progress of their cake through a simple online tracking page. The evaluation demonstrated that clear status tracking improved kitchen efficiency, reduced customer inquiry calls, and ensured timely order fulfillments during busy operations.

This local study directly informs the order management module of the proposed platform, highlighting the necessity of step-by-step order tracking from kitchen preparation to final delivery. Without clear visual status updates, kitchen staff can easily mismanage prep priority. The proposed system will incorporate a structured workflow tracker based on the findings of Aquino et al., enabling the bakery manager to update prep stages seamlessly and keep customers informed about their cake's real-time progress.

The rapid shift toward digital payments among local micro, small, and medium enterprises provides necessary context for modern e-commerce systems. Soriano et al. (2023), in *E-Commerce Adoption and Digital Payment Integration Among Small Food Enterprises in Metro Manila* (Research Congress on Philippine Business and Information Technology, ISSN: 2546-0617), evaluated the adoption rate of digital payment channels among small food vendors and bakeries in Metro Manila. The authors explained that local consumers increasingly expect small food businesses to

offer cashless payment methods, such as GCash and Maya, alongside traditional cash transactions.

The researchers analyzed businesses that integrated automated e-wallet payment options into their ordering platforms against those that relied solely on manual bank transfers or cash on delivery. Their findings confirmed that integrating digital payment methods significantly increased completed checkout rates, reduced unpaid reservation cancellations, and simplified daily cash reconciliation for business owners. Offering instant digital payment channels gave customers greater confidence and convenience during checkout.

This study provides direct justification for integrating e-wallet payment verification into the proposed platform. Custom cake orders require deposit payments before baking begins to avoid wasted ingredients from bogus buyers. By building upon the digital payment insights of Soriano et al., the proposed system will include convenient e-wallet payment verification, ensuring that local cake buyers have a seamless, secure, and convenient checkout experience while protecting the bakery from bogus reservations.

2.3 RELATED LITERATURE

Academic literature on local software frameworks highlights the structural requirements necessary for building automated bakery order platforms. Perez and De Leon (2022), in *Frameworks for Automated Web-Based Ordering and Operational Management in Small Food Enterprises* (Philippine Information Technology Education Journal, ISSN: 2423-1126), evaluated the structural software standards required for specialized food ordering applications. The authors emphasized that generic e-commerce templates fail to address the specific operational constraints of local bakeries. Standard online shopping carts are designed for pre-packaged, ready-to-ship items and lack built-in logic for tracking custom cake specifications, calculating dynamic prep times, or managing daily baking limits.

Their literature review established that integrating custom order configuration, real-time schedule locking, and visual revenue reporting into a single web application creates a comprehensive solution for small-scale food enterprises. The authors outlined functional specifications for multi-tier selection tools, administrative schedule management, and automated sales graphing. Their framework demonstrated that combining these operational modules under one system improves

order management efficiency while providing small shop owners with essential business control.

This literature serves as a primary structural benchmark for the current project, establishing the exact core modules needed to address local operational challenges. By adopting the software framework established by Perez and De Leon, the proposed platform will integrate a custom cake builder, a smart scheduling calendar, and a sales analytics dashboard into a unified web application. This ensures that the platform addresses the specific operational needs of local Philippine bakeshops rather than relying on inadequate generic e-commerce templates.

National policy reports and economic frameworks emphasize the urgent need for digital adoption among Philippine micro, small, and medium enterprises. The World Bank Group and the Department of Trade and Industry (2022), in *Digitalization and E-Commerce Adoption Among Micro, Small, and Medium Enterprises (MSMEs) in the Philippines* (World Bank Group & DTI Joint Policy Report, ISBN: 978-971-9123-45-6), evaluated the digital readiness of small businesses across the country. Their publication noted that over ninety percent of registered food enterprises in the Philippines operate as MSMEs, with many still relying on manual bookkeeping and traditional face-to-face transactions.

The report highlighted that micro-enterprises adopting digital channels experience higher operational efficiency, improved market reach, and greater resilience against economic disruptions. However, the report also identified major barriers to technology adoption, including the complexity of existing software platforms and high subscription costs for small shop owners. The authors advocated for simple, accessible, and localized web platforms tailored to specific enterprise needs, ensuring that small business owners can operate them without expensive technical setup costs.

This official literature provides macroeconomic justification for the current project, proving that digital transformation is essential for local bakeries to maintain profitability and competitiveness in the modern economy. Small food businesses cannot thrive on manual methods alone in an increasingly digital market. By following the policy recommendations of the World Bank Group and DTI, the proposed platform will offer a simple, accessible, and localized web solution that helps community bakeries transition to digital e-commerce without facing steep software learning curves or high financial barriers.

Regulatory frameworks from central financial institutions further underscore the importance of integrating modern e-wallet solutions into local e-commerce systems. The Bangko Sentral ng

Pilipinas (2023), in *Digital Payments Transformation Roadmap and the Growth of E-Wallets in Small Retail Operations* (BSP Policy Framework Series, ISSN: 2719-0129), outlined the national initiative to transition Philippine retail commerce toward a cash-lite economy. The BSP documented a rapid increase in consumer reliance on digital wallets like GCash and Maya, noting that local shoppers now prefer electronic payment options for daily purchases and online food orders.

The framework emphasized that small food businesses incorporating direct e-wallet payment options experience fewer cancelled reservations, faster payment settlements, and simplified financial tracking. Relying exclusively on cash on delivery or manual bank transfers creates transaction friction and financial risk for micro-enterprises. The report stressed that modern web applications must accommodate local payment gateways to meet changing consumer habits, streamline financial audits, and support national financial inclusion goals.

This literature provides direct regulatory backing for the proposed system's payment integration feature. Custom cake orders require advance deposits to guarantee order commitments and prevent financial losses from uncollected items. By aligning with the national digital payments roadmap established by the Bangko Sentral ng Pilipinas, the proposed platform will

integrate convenient e-wallet verification options, giving local customers a familiar checkout method while ensuring secure reservation deposits for the bakery owner.

Technical literature on modern database design explains the necessity of cloud synchronization in interactive web applications. Rogers et al. (2021), in *Architectural Principles of Cloud-Based NoSQL Databases and Real-Time Synchronization in Web Applications* (ACM Digital Library / Academic Press, ISBN: 978-1-4503-8000-3), detailed the structural advantages of using cloud-hosted NoSQL databases for dynamic e-commerce platforms. The authors explained that traditional relational databases often experience performance lag or indexing delays when handling rapid, concurrent real-time updates from multiple simultaneous users.

In contrast, non-relational document databases allow continuous, multi-user data synchronization without server slowdowns or system crashes. When a customer modifies a custom order specification or books a delivery date on the frontend web interface, the database instantly broadcasts that update to the admin dashboard. The research confirmed that real-time document-based syncing maintains data integrity, reduces server load, and delivers an instant response across multi-device administrative setups.

This literature provides essential technical justification for selecting a real-time cloud database architecture for the proposed system. A cake ordering platform requires constant data exchange between frontend customer inputs and backend kitchen dashboards. Applying the cloud architectural principles outlined by Rogers et al. ensures that customer orders, custom cake selections, and schedule slot locks update instantaneously across the proposed platform without requiring manual web browser page reloads.

Human-computer interaction guidelines provide clear visual standards for designing intuitive product configurators in e-commerce applications. Martin et al. (2020), in *Human-Computer Interaction Guidelines for Custom Product Configurator Interfaces in E-Commerce* (IEEE Transactions on Human-Machine Systems, ISSN: 2168-2291), examined interface usability for customized ordering platforms. The authors explained that non-technical buyers experience decision fatigue and confusion when confronted with overly complicated, unstructured order forms filled with excessive text boxes and confusing choices.

To optimize user interaction, the authors recommended implementing step-by-step wizard interfaces, clear visual selection cards, dynamic cost updates, and explicit input fields. Their research proved that intuitive configurator

designs increase order completion rates, reduce user input errors, and significantly improve overall user satisfaction. Structuring custom choices logically helps users complete complex orders quickly without feeling overwhelmed.

This technical literature directly informs the visual layout of the proposed system's cake customization page. Ordering a custom cake involves selecting multiple variables, such as size, flavors, frosting styles, and decorations. By applying the human-computer interaction principles recommended by Martin et al., the proposed platform will feature a clean, user-friendly cake builder that guides local customers through the customization process effortlessly, ensuring an enjoyable user experience while capturing accurate order details.

2.4 SYNTHESIS OF THE STUDY

The literature and studies reviewed in this chapter collectively emphasize the crucial role of digital transformation, specialized custom ordering workflows, dynamic calendar scheduling, and visual sales analytics for small food and retail enterprises. Both foreign and local researchers agree that traditional, paper-based record-keeping or unorganized social media messaging methods create severe operational bottlenecks for growing bakeries. Relying on physical notebooks, verbal kitchen handoffs, or static messaging threads frequently leads to lost order slips, delayed production, misplaced customer preferences, and significant administrative fatigue for small bakery owners.

Foreign studies demonstrated the technical feasibility and efficiency of modern web architectures in eliminating these manual administrative bottlenecks. Liang et al. (2021) and Mali (2020) proved that integrating real-time cloud databases and separating off-the-shelf inventory from custom order pipelines drastically reduces administrative processing delays and prevents duplicate order entries. Furthermore, Rajaput et al. (2022) established that interactive, step-by-step cake customization builders eliminate buyer-baker miscommunication by capturing design specifications upfront, while Sharma et al.

(2020) highlighted the necessity of dynamic capacity allocation algorithms to prevent kitchen staff burnout caused by order overbooking. In terms of business management, Kumar et al. (2021) showed that visual sales analytics dashboards allow non-technical store owners to quickly interpret revenue trends and make informed inventory purchasing decisions without sifting through complex accounting ledgers.

Local studies provided essential context regarding the specific operational realities, structural constraints, and digital readiness of bakeries across the Philippines. Research from the Neliti Repository (2019) on Manolette Bakeshop, along with studies by Santos et al. (2021) in Laguna and Garcia et al. (2022) in Cavite, confirmed that local MSME bakeries frequently struggle with misplaced paper order slips, ingredient shortages, and excessive manual messaging on social media apps. Furthermore, Aquino et al. (2020) demonstrated that structured, step-by-step order status tracking improves kitchen productivity in Central Luzon bakeries, while Soriano et al. (2023) highlighted the rapid adoption and necessity of integrating e-wallet payment options like GCash and Maya to secure reservation deposits for food enterprises in Metro Manila.

The related literature reinforced these empirical findings from institutional, regulatory, technical, and human-computer

interaction perspectives. Academic literature by Perez and De Leon (2022) established that standard off-the-shelf e-commerce templates are inadequate for custom cake operations because they lack specialized customization logic and dynamic schedule-locking capabilities. Policy reports from the World Bank Group and DTI (2022), alongside national digital payment frameworks from the Bangko Sentral ng Pilipinas (2023), provided macroeconomic and regulatory justification for digitalizing local food enterprises through accessible e-commerce and cashless payment channels. Finally, technical literature by Rogers et al. (2021) on cloud NoSQL database synchronization and design guidelines by Martin et al. (2020) on human-computer interaction supplied the theoretical foundation for building an intuitive, real-time web application tailored to small business operations.

The Research Gap

Despite the extensive research on bakery management, e-commerce adoption, and custom order platforms, a significant operational gap remains for local community bakeries in the Philippines. Existing foreign and local systems usually address these features in isolation—offering either a standalone cake builder,

a basic inventory tool, or a generic appointment booking calendar. Most commercially available platforms are either too expensive for local micro-enterprises, overly complicated to set up, or lack dynamic scheduling logic that automatically locks calendar dates based on daily kitchen labor capacity.

Furthermore, generic ordering templates rarely combine custom cake configuration, dynamic slot blocking, local e-wallet payment verification, and visual sales analytics into a single, unified web platform tailored to local Philippine bakeshop operations.

Additionally, most established e-commerce solutions cater primarily to large-scale retail franchises with standardized product lines. Micro and small bakeries in provincial communities operate under unique labor constraints, where a single master baker or small kitchen team must balance daily walk-in bakery sales with highly detailed, labor-intensive custom cake orders. When existing platforms fail to account for daily kitchen labor thresholds or lack localized payment verification methods like GCash, local bakers are forced back into manual social media messaging threads, re-exposing their operations to lost orders, overbooking, and uncollected reservations.

Contribution of the Proposed Study

The proposed platform addresses this research gap by integrating these essential operational tools into one accessible, web-based system designed specifically for local Philippine bakeshops. By synthesizing the architectural models of Liang et al. (2021) and Mali (2020), the proposed system will utilize a real-time cloud database to keep customer orders and kitchen management panels synchronized instantly. Adopting the step-by-step customization approach of Rajaput et al. (2022) and the user interface guidelines of Martin et al. (2020), the platform will feature an intuitive online cake builder that captures complete order specifications—such as tiers, flavors, frosting styles, and reference image uploads—upfront during checkout.

To solve the issue of operational overbooking highlighted by Sharma et al. (2020), the proposed system will implement smart scheduling logic that automatically calculates daily baking capacity and locks checkout slots when maximum limits are reached. Building upon the workflow tracking model of Aquino et al. (2020), the system will provide an administrative pipeline tracker so kitchen staff can manage order progress seamlessly from prep to pickup. Additionally, building upon the local findings of Soriano et al. (2023) and the BSP (2023) payment roadmap, the platform will integrate local e-wallet options to

ensure smooth, cashless deposit transactions. Finally, drawing from the data visualization principles of Kumar et al. (2021) and the framework by Perez and De Leon (2022), the system will provide local bakery managers with an easy-to-read sales analytics dashboard to track income trends and top-selling products effortlessly.

CHAPTER 3

This chapter presents the research design, system development methodology, system architecture, tools and technologies, data gathering techniques, testing plans, and evaluation metrics used to design, code, and deploy the web platform for C&N Bakery on Vercel.

3.1 Research Design

In this study, the research team uses a descriptive-developmental research design. This practical, two-part approach combined two distinct phases:

- **The Descriptive Phase:** Observing, studying, and documenting how C&N Bakery managed its daily operations using handwritten logbooks, wall calendars, loose paper slips, and physical paper receipts.
- **The Developmental Phase:** Planning, coding, testing, and refining a web-based ordering and kitchen management platform hosted and deployed on Vercel to solve those specific operational struggles.

By linking these two phases, the research team ensured that every button, database table, form input field, and dashboard

tool will address a real problem the bakery faces every single day, rather than making guesses about what staff might need.

DESCRIPTIVE-DEVELOPMENTAL RESEARCH DESIGN

DESCRIPTIVE PHASE	DEVELOPMENTAL PHASE
• Observed walk-in & phone orders • Documented paper logbook issues • Identified wall calendar risks • Recorded manual math errors	System Infrastructure & Security • Cloud database, Vercel deployment & email alerts Customer Portal & Ordering System • Custom cake builder & daily order limits Admin & Kitchen Management System • Sales analytics & dynamic menu CRUD

Figure 3.1: Descriptive-Developmental Research Methodology Framework

3.1.1 Descriptive Research

The descriptive phase focused on observing and documenting how C&N Bakery operated in its natural daily environment. Rather than setting up artificial tests or altering store routines, the

research team spent time inside the bakery to study how orders were taken, processed, tracked, and reconciled.

Application in the Study

Before writing any software code, the research team visited C&N Bakery to examine its manual workflow. The team interviewed the business owner, spoke with counter staff, observed kitchen operations, and reviewed physical order records. Through these field observations, four major operational challenges were identified:

1. Disorganized Customer Messages and Order Notes

- **Background:** Customer inquiries and custom cake requests came in through different channels – personal phone calls, text messages (SMS), and chat threads on the C&N Bakery Facebook page – with no single place to keep track of them.
- **Operational Impact:** When a customer called to check on an order or adjust a pickup schedule, staff had to stop helping walk-in customers to scroll through long message threads on personal smartphones just to locate reference photos, agreed price quotes, or pickup notes.

- System Solution: A centralized Customer Portal will store order histories, design preferences, and communication notes in a single cloud database.

2. Risks of Handwritten Paper Order Slips

- Background: Custom cake choices such as layer dimensions, flavor choices, frosting types, color themes, and written dedications were recorded by hand on loose paper slips.
- Operational Impact: Fast handwriting recorded during busy shop hours was often hard for kitchen staff to read, leading to mistakes in cake text spellings or icing colors. In addition, paper slips were easily misplaced or damaged near kitchen prep areas.
- System Solution: Digital order forms will capture customer choices through standardized inputs and image reference uploads, making details clear and permanently recorded.

3. Manual Scheduling and Kitchen Overbooking

- Background: Delivery and pickup dates relied entirely on a physical paper wall calendar hanging near the kitchen prep station.
- Operational Impact: The kitchen team could safely bake and decorate 3 to 5 custom cakes per day without

compromising product quality. Counter staff taking phone orders during busy shop hours frequently forgot to check the wall calendar before accepting new custom orders, causing accidental double-bookings and delayed deliveries.

- System Solution: An automated booking calendar will track daily order volume and automatically close dates that reach maximum kitchen capacity.

4. Manual Bookkeeping and Calculation Errors

- Background: Tracking customer downpayments, cross-checking online bank transfer screenshots, and calculating daily sales totals were done using handheld calculators and physical logbooks.
- Operational Impact: Calculating sales totals by hand after long working shifts increased the risk of simple math mistakes. Misread numbers or misplaced receipts forced the store owner to spend over an hour re-checking individual paper slips at closing time.
- System Solution: An automated sales analytics dashboard will calculate revenue trends, track payment statuses, and generate financial reports automatically.

3.1.2 Developmental Research

Developmental research focuses on using real field observations from the descriptive phase to design, construct, test, and refine a practical software solution. Instead of building generic features, every tool and module in the proposed platform addresses a documented operational problem at C&N Bakery.

Application in the Study

During the developmental phase, the researchers will translate field observations into working software components across three core development areas, deployed via Vercel's cloud application platform:

Table 3.1: Developmental Phase Module Implementation Blueprint

Development Area	System Module Scope	Planned Technical Features & Capabilities
1.System Infrastructure & Security	Cloud Database, Access Control & Vercel Deployment	• Will design cloud database schemas to store customer profiles, product catalogs, and order records securely.

		• Will implement Role-Based Access Control to separate administrative permissions from customer privileges. • Will configure automated email triggers corresponding to order confirmation, downpayment verification, and delivery/pickup status. • Will deploy the application on Vercel for fast, reliable cloud access.
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2. Customer Portal & Ordering System	Customer Storefront & Interactive Checkout	• Will develop catalog browsing for guest users. • Will build an interactive Custom Cake Builder with real-time price calculations. • Will enforce pickup/delivery date checks and daily order limits during checkout. • Will support COD, GCash QR, and Bank Transfers with Facebook receipt checks. • Will build a visual Order Tracker with real-time
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		status steps.
3. Admin & Staff/Kitchen Management Dashboard	Business Analytics & Store Control	• Will construct a Sales Analytics dashboard to summarize revenue trends. • Will implement a dual-section Order Dashboard (New vs. Active orders). • Will build an interactive Schedule Calendar to toggle daily limits. • Will provide dynamic CRUD controls for menu items, prices, and cake choices.

		• Will maintain an automated history log for all system updates.
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3.2 System Development Methodology

For this study, the research team selected the Agile Development Model as the primary framework for the Software Development Life Cycle (SDLC). Traditional software approaches, such as the Waterfall model, follow a rigid sequence where each phase must be finished completely before the next can begin. Under Waterfall, making changes to earlier choices after coding starts is difficult and expensive.

Developing a web-based ordering and management platform for a small business like C&N Bakery required flexibility. As the business owner, kitchen staff, and test customers interact with early site builds, practical operational details inevitably arise that were not anticipated during initial planning.

The Agile methodology will resolve this by dividing project development into short, manageable iterations called sprints. Instead of waiting until the end of the project to test the

entire application, the bakery owner and staff will review working feature releases early and provide direct feedback. This continuous feedback loop will reduce technical risks, prevent unnecessary feature development, and ensure the final software fits seamlessly into daily bakery operations.

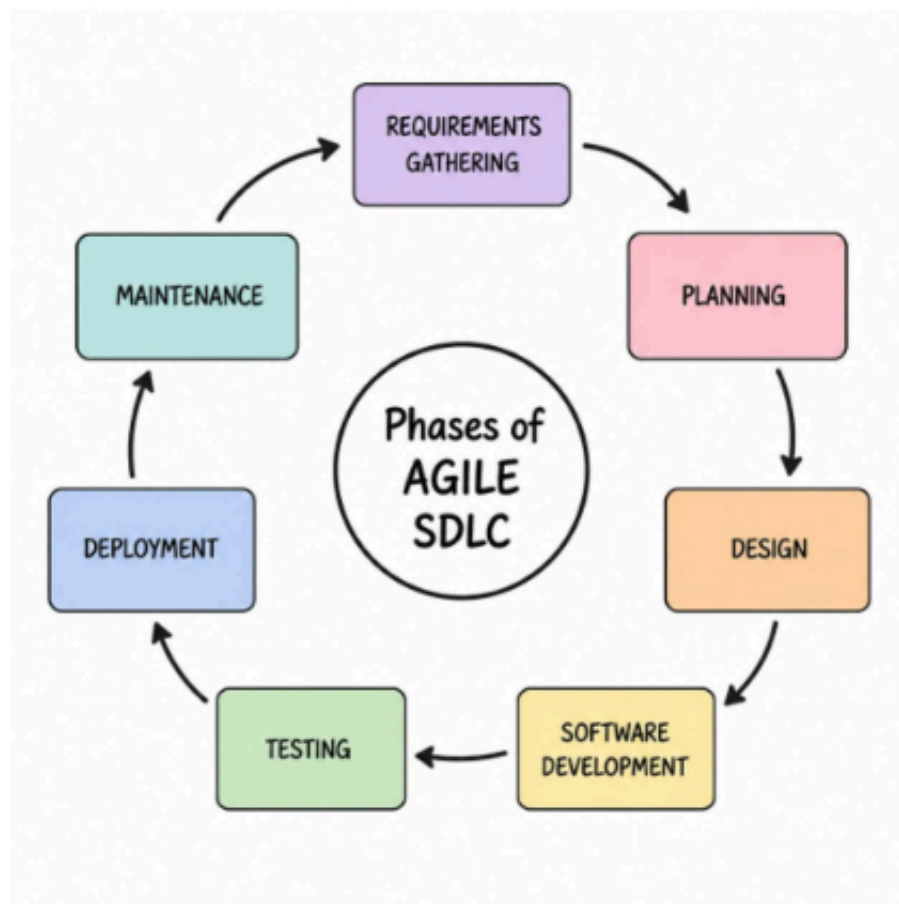


Figure 3.3: 7-Phase Agile SDLC Methodology Flow

3.2.1 Requirements Gathering

Concept & Purpose

The requirements gathering phase will focus on identifying the exact features, operational rules, and performance targets needed for the web platform. Setting clear feature limits early will keep development on schedule and prevent adding unnecessary tools halfway through the project.

Application to C&N Bakery Platform

During field visits to C&N Bakery, the research team interviewed the owner and shop staff to separate essential tools from features that would add unnecessary complexity. To keep the project realistic and manageable, advanced corporate tools—such as live GPS delivery tracking and automated inventory reordering—were excluded from the project scope.

From these discussions, the team grouped the system requirements into two main categories:

- **Functional Requirements:** The 8 core system tools (Account Sign-Up & Browsing, Custom Cake Builder, Customization Details, Cart & Progress Tracker, Smart Capacity Calendar, Admin Management Portal, Email Notifications, and Sales Analytics).

- **Non-Functional Requirements:** Operational standards defining site speed, system security, and mobile responsiveness.

Summary List of Functional Features

1. **Account Sign-Up & Browsing:** Allows customers to browse products as guests or create accounts to save addresses and view past orders.
2. **Interactive Custom Cake Builder:** Calculates cake order prices automatically as buyers select tiers, sizes, chiffon flavors, and frosting options.
3. **Order Customization Details:** Input fields for cake dedications, frosting color choices, and external design reference links.
4. **Cart & Order Progress Tracker:** A digital cart paired with a real-time progress tracker (Preparing, Baking, Packing, Ready).
5. **Smart Capacity Calendar:** Checks kitchen daily order limits and locks out fully booked pickup or delivery dates.
6. **Admin Management Portal:** A secure dashboard for bakery staff to review custom orders, approve payments, and print daily bake lists.
7. **Email Notification System:** Automatically emails customers whenever staff update their order status, payment verification, or fulfillment schedule.

8. Sales Analytics Dashboard: Displays simple visual charts showing daily, weekly, and monthly store revenue.

Table 3.1: Detailed Functional Requirements Matrix

Feature Name	User Role	Planned System Actions	Input and Output Details
Account Sign-Up & Browsing	Guest / Customer	Users will create accounts using their email addresses or browse products as guests. Signed-in accounts will store saved addresses and past orders.	Input: User name, email, contact number. Output: Account profile and login session.
Interactive Custom Cake Builder	Customer	The site will recalculate prices	Input: Chosen tiers, cake sizes, flavor

		automatically as buyers pick cake tiers, sizes, chiffon flavors, frosting options, and upload design screenshots.	selections, image uploads. Output: Calculated custom cake price total.
Customization Details	Customer	The interface will provide text boxes for custom cake dedications, frosting color choices, and links to reference photos.	Input: Dedication text, color selections, reference links. Output: Saved order customization details.

Cart & Order Tracking	Customer	Customers will manage chosen items inside a digital shopping cart and monitor order progress through real-time step indicators (Preparing, Baking, Packing, Ready) .	Input: Cart items, selected order ID. Output: Itemized cart summary and live order progress tracker.
Smart Capacity Calendar	Customer	The system will check daily kitchen capacity records before confirming	Input: Selected pickup or delivery date.

		pickup or delivery dates. If a date hits its limit, the calendar picker will block new bookings.	Output: Date availability status (Available or Fully Booked).
Admin Portal	Admin /Staff	Kitchen staff will log into a secure dashboard to review orders, approve custom requests, print baking lists, and update order statuses.	Input: Admin credentials, status update toggles. Output: Filtered order dashboard and printable bake lists.

Email Notifications	System / Customer	The system will send automatic email updates regarding order confirmations, downpayments, and delivery/pick up statuses whenever kitchen personnel update an order.	input: Database order status trigger. Output: Automated customer email update.
Sales Analytics Dashboard	Admin	The dashboard will display simple visual charts	Input: Date range filters, paid transaction

		summarizing daily, weekly, monthly, and custom date range revenue totals.	records. Output: Visual sales totals and financial summary graphs.
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Table 3.2: Operational Performance

Requirement Category	Performance Target	How the System Will Achieve It
Responsive Layout	Web pages will adjust cleanly across desktop computers, tablets, and mobile phones.	Front-end layouts will be built using CSS fluid grids and responsive design.
System Security	Admin management tools will be	Staff routes will require encrypted

	protected from unauthorized access.	logins and Role-Based Access Control (RBAC).
Speed & Performance	Page load speeds will remain under fast response thresholds on both Wi-Fi and mobile data	Front-end code will be optimized, uploaded pictures will be compressed, and assets will be served via Vercel's global Content Delivery Network (CDN).

3.2.2 Planning

Concept & Purpose

The planning phase will break down system goals into manageable coding tasks and schedule them across specific development sprints. It will establish software tools, cloud backend environments, and budget constraints early to keep development organized and on schedule.

Application to C&N Bakery Platform

To keep the website fast, reliable, and free to host, the research team planned a hybrid cloud architecture combining Vercel Platform Hosting for front-end web delivery with Google Firebase Cloud Services for backend services. Because local bakeries usually operate on tight budgets, keeping operational running costs at zero was set as a primary design goal. By leveraging free developer tiers across both platforms, the platform stays zero-cost through three main strategies:

- **Free Web Hosting:**The website's front-end code will be deployed live using Vercel's global Content Delivery Network (CDN), providing fast rendering and built-in SSL security without hosting fees.
- **Zero-Cost Image Storage:** Firebase Realtime Database and Authentication will safely store customer orders, login accounts, and uploaded custom cake reference images under Firebase's free Spark plan.
- **Manual Payment Verification:** To avoid paying third-party payment gateway fees, customers will pay via GCash QR codes or direct bank transfers and attach receipt screenshots. Bakery staff will verify these screenshots in the FB Page of CandN bakery before confirming the order.

Summary of Selected Software Stack

The development team selected the following key software tools based on speed, ease of use, and generous free developer limits:

- **HTML5:** Provides semantic web structure, accessible form fields for cake customization, and cross-browser markup standard across mobile and desktop browsers.
- **CSS3:** Builds responsive, fluid UI layouts and forms that adjust cleanly across smartphones, tablets, and desktop displays.
- **JavaScript (ES6 OOP):** Organizes website functions into reusable, secure code blocks called classes. It uses private data protection (Encapsulation), simple backend commands (Abstraction), shared card blueprints (Inheritance), and dynamic button behaviors (Polymorphism) to keep the system modular and easy to manage.
- **Vercel Hosting:** Deploys and serves the web application front-end interface live using Vercel's edge network for optimal loading speeds.
- **Firebase Realtime Database:** Syncs order updates instantly between customer tracking screens and the kitchen admin dashboard without requiring page refreshes.

- **Firebase Authentication:** Handles user account logins, password security, and staff administrative permissions safely.

Table 3.3: Software Stack & Zero-Cost Technical Justification

Development Component	Tool / Technology Selected	Why the System Will Use This	Free Tier & Budget Strategy
Markup & Page Layout	HTML5	Will deliver clean, semantic DOM structures and standardized input fields for custom cake forms.	Native browser standard; will require zero licensing fees or external dependencies.
Client-Side Application Logic	JavaScript (ES6 OOP)	Will structure reusable browser logic into classes applying Encapsulation, Abstraction,	Will run natively in all web browsers without requiring paid server-side

		Inheritance, and Polymorphism.	runtime environments.
Web Hosting	Vercel Hosting	Will host and deliver web platform pages fast and securely via Vercel's CDN	Will remain completely free on Vercel's Hobby Plan with generous monthly bandwidth limits.
UI Styling & Responsiveness	CSS3	Will build interactive forms and product grids that load fast and look clean on mobile devices.	Will compile directly into static CSS files hosted for free on Firebase Hosting.
Database Dashboard	Firebase Realtime DB	Will sync new order alerts	Will utilize free daily

		to kitchen staff screens instantly as soon as a customer submits an order.	database traffic suitable for daily local bakery operations.
User Access & Security	Firebase Authentication	Will encapsulate login sessions, password security, and staff access permissions behind clean wrapper methods.	Will remain free for active user logins every month.

3.2.3 Design

Concept & Purpose

The design phase converts feature lists into visual blueprints and structural layouts before active coding begins. Creating wireframes early allows layout adjustments on paper, saving development time during the coding phase.

Application to C&N Bakery Platform

Visual wireframes will be drawn for both the customer-facing storefront and the admin dashboard. Database structures will be outlined in the cloud database to maintain organized records:

- **Order Schema Structure:** Cake selections, design notes, image links, pickup dates, and payment statuses will link directly to a single, unique Order ID.
- **Calendar Capacity Rules:** Database tables will track active custom bookings for each date, comparing incoming requests against daily limits and locking out dates that reach maximum capacity.
- **Sales Data Schema:** Transaction totals will be stored with clear payment status tags, allowing the sales analytics dashboard to process financial metrics without slowing down the site.

3.2.4 Development

Concept & Purpose

The development phase will involve writing code, building user screens, connecting cloud backend services, and turning wireframe designs into working software. Under our Agile setup, development will be divided across three focused coding cycles called sprints.

Application to C&N Bakery Platform

Development will be completed across three sequential sprints, with each sprint focusing on specific platform modules:

1. Sprint 1: Customer Storefront & Custom Cake Builder

During Sprint 1, the research team will focus on front-end user interfaces. HTML5 markup, CSS3, and JavaScript will be written to create the user storefront. Key tasks will include building responsive product grids, coding the guest checkout workflow, and creating the interactive Custom Cake Builder with real-time price calculations.

2. Sprint 2: Admin Dashboard & Kitchen Operations Panel

Sprint 2 will focus on administrative and management tools for C&N Bakery staff. The team will build secure staff login routes, the dual-section Order Dashboard (New vs. Active orders), and interactive schedule controls. Code

logic will be programmed to track daily order limits per date, automatically closing dates that hit maximum kitchen capacity. Simple visual charts will also be built for the sales analytics module.

3. **Sprint 3: Cloud Backend Integration, Database Security, & Vercel Deployment**

In Sprint 3, the front-end user interfaces will be connected directly to cloud database services and deployed via Vercel. The team will bind online form fields to cloud database paths, set up customer authentication, configure image uploads for payment receipts, and apply Role-Based Access Control security rules. Automated email triggers will also be set up to send status update alerts to customers.

Table 3.6: Agile Sprint Development

Sprint Phase	Main Coding Focus	Specific Coding Tasks	Deliverables Produced
Sprint 1	Customer Storefront & Cake Builder	• Will build user sign-up and login forms.	Working storefront, interactive cake customization

		• Will create product catalog grids. • Will code interactive Custom Cake Builder price calculations. • Will implement digital shopping cart management.	form, and functional shopping cart screens.
Sprint 2	Admin Management Dashboard	• Will build secure staff login screens and role authorization •	Functional admin management dashboard, interactive schedule calendar, and sales

		• Will program daily capacity calendar control rules. • Will build order management lists and sales analytics charts. • Will provide dynamic CRUD tools for menu pricing and cake options.	reporting interface.
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Sprint 3	Cloud Integration, Vercel Deployment & Live Updates	• Will connect JavaScript user forms to cloud database paths. • Will set up dynamic email notifications for order confirmation, downpayment, and delivery/pick up triggers. • Will configure cloud storage for receipt and cake image uploads.	Fully connected, live web platform ready for multi-stage system testing.
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		• Will apply role-based database security rules and deploy live on Vercel.	
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3.2.5 Testing

Concept & Purpose

The testing phase will verify that all web features work correctly, remain secure, and run smoothly across different devices before public deployment. In an Agile methodology, testing will happen continuously after each development sprint to catch and fix bugs early.

Application to C&N Bakery Platform

The research team will execute a multi-stage testing plan across four levels:

1. Unit Testing: Testing individual code functions independently (such as automated price calculator math and form input checks).
2. Integration Testing: Making sure data flows accurately between online forms, user profile accounts, and cloud database nodes.
3. End-to-End System Testing: Running complete trial test orders from start to finish—from initial cake customization to admin order approval and sales logging.
4. User Acceptance Testing (UAT): Holding hands-on trial sessions with key C&N Bakery stakeholders (the owner, bakers, counter staff, and test customers) to confirm the system is easy to use.

3.2.6 Deployment

Concept & Purpose

Deployment will involve publishing the fully tested web application to live cloud hosting platforms, setting up production database paths, and preparing the software for daily business use.

Application to C&N Bakery Platform

The application will be published live using Vercel cloud hosting linked directly to cloud database services. Before opening the site to real customers, the research team will clear all test orders, sample photos, and fake accounts from the database. The live database will then be loaded with C&N Bakery's official product menu, correct prices, product categories, and daily kitchen capacity limits.

3.2.7 Maintenance

Concept & Purpose

Following deployment, the maintenance phase will focus on keeping track of site performance, collecting user feedback during daily shop operations, and making minor adjustments to ensure system stability.

Application to C&N Bakery Platform

A post-launch maintenance plan will be implemented:

- **Performance Checks:** Database response speeds, Vercel serverless function logs, and automated calendar date locks will be checked regularly.
- **Staff Feedback Logbook:** A physical feedback logbook will be placed in the bakery kitchen for staff to write down daily

operational notes or minor issues encountered during shop hours, ensuring ongoing software stability and ease of use.

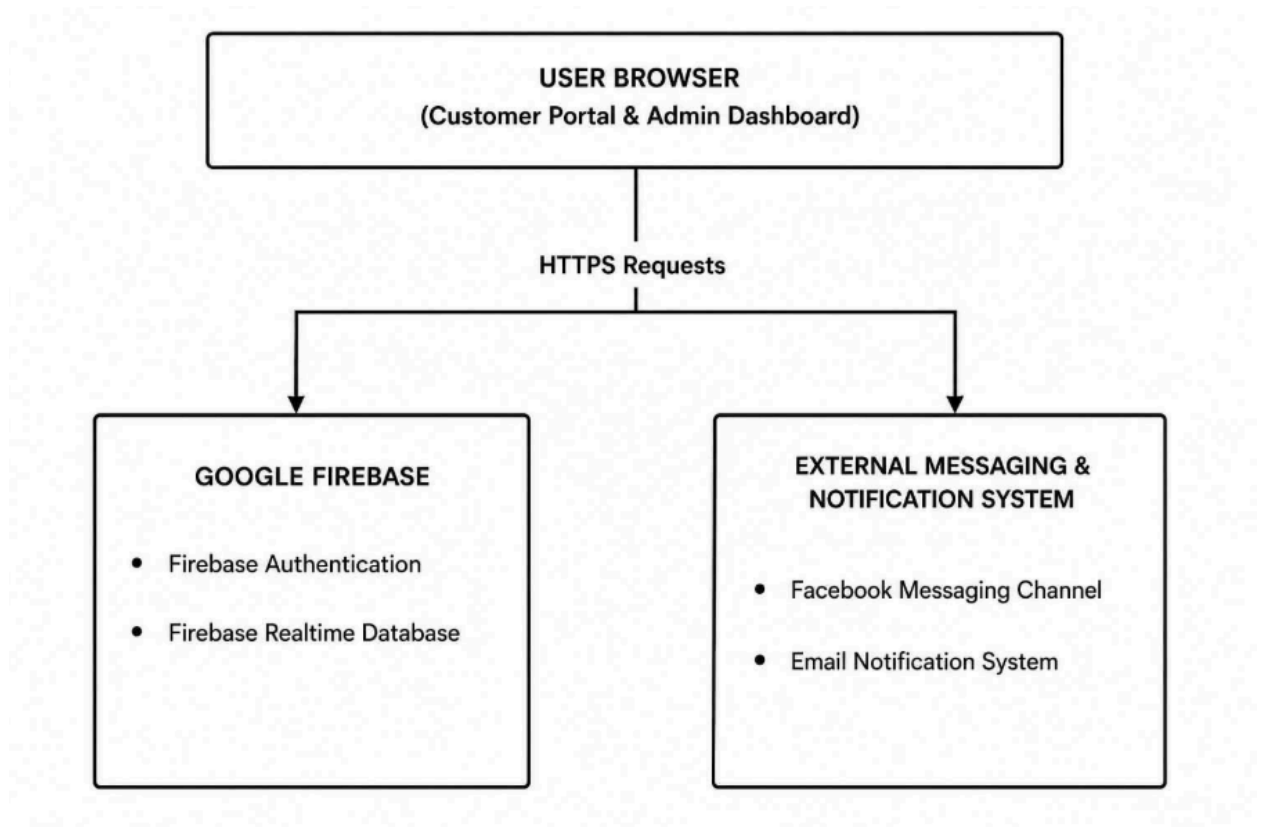
3.3 System Architecture

The web application will use a cloud-based software setup deployed live on Vercel for front-end web hosting, paired directly with Google Firebase backend cloud services. Instead of building and maintaining an old-school physical web server like running Apache with PHP and an offline database, this modern cloud setup lets our web pages communicate directly with cloud services over secure HTTPS connections.

Adopting a cloud-first architecture minimizes setup time, eliminates physical server maintenance, and avoids monthly hosting fees. To keep system processing lightweight and fast, data is categorized into two distinct types:

- **Lightweight Text Data:** Customer profiles, catalog items, custom cake options, daily kitchen limits, order records, and site settings will stay stored as simple JSON text nodes inside the Firebase Realtime Database.
- **Media & File Storage:** Custom cake design reference photos will be stored safely in Firebase. Payment receipt

screenshots are managed manually through C&N Bakery's official Facebook Messenger page to avoid third-party gateway.



3.3.1 User Browser & Web Application Layer

The User Browser layer will serve as the visual front-end of the web application. It will include all the pages, buttons, forms, and product menus that customers and staff will see and interact with on a phone, tablet, or computer browser.

To maintain security and a clean user experience, the system will separate front-end access into two distinct web interfaces:

- **Public Customer Portal:** The main website accessible to standard customers for browsing products, building custom cake orders, tracking purchases, and creating customer user accounts.
- **Admin Portal:** A separate web address/link reserved strictly for bakery staff and the system administrator to manage products, adjust daily kitchen limits, and process live customer orders.

The front-end screens for both portals are designed using standard web technologies: HTML5 for page layouts, CSS3 for mobile-responsive styling, and JavaScript for user interactions. The user browser will handle interactive features such as the Custom Cake Builder, category filtering, and automatic price calculations right on screen. Running calculations directly in the browser will give customers instant price updates as they pick cake sizes, flavors, and design choices without making them wait for page reloads.

Using cloud services allowed the web app to connect directly to ready-made backend tools instead of building and running a physical server from scratch. Google Firebase will serve as the

primary cloud platform for the application, handling background operations such as user account checks, real-time data saving, and database security.

Google Firebase will manage platform operations through two core cloud modules:

Component	Function	Description
Vercel Hosting	Front-End Delivery	Will host the web application and serve screens fast to users over secure HTTPS.
Firebase Authentication	Identity & Access	Will manage customer account sign-ups on the main site, as well as staff and admin credentials on the dedicated Admin Portal link.
Firebase Realtime Database	Live Data & Image Storage	Stores structured JSON records (users, products,

		categories, customization-options, orders, dayLimits, dayFulfillment, carts, settings) and image URL links, syncing updates live across active client sessions.
External Integrations	Customer Communication	Will handle automated status emails and direct customer chats on Facebook Messenger.

1. Firebase Authentication

This module will act as the primary security layer for both web entry points. On the public Customer Portal, it will manage standard user account sign-ups and logins. On the separate, dedicated Admin Portal link, it will authenticate staff credentials and verify administrative privileges. Whenever an account attempts to open restricted administrative endpoints or

modify system menu settings, Firebase Authentication will verify their account access level. If an unauthorized user or standard customer attempts to access the dedicated admin URL directly, the system will block access right away and redirect them back to the customer home page.

2. Firebase Realtime Database

This module will act as the central storage vault for the platform. It will hold structured JSON records and product image links across dedicated root nodes: users for customer and staff account details, products and categories for catalog items and cake photos, customization-options for cake builders, orders for active purchases, and dayLimits with dayFulfillment for kitchen scheduling. Because it syncs data instantly using WebSockets, whenever a staff member updates an order status on the Admin Portal, that change will appear right away on the customer's order tracking screen without requiring a manual page refresh. It will also manage kitchen capacity by counting active bookings under dayFulfillment for a chosen date and automatically blocking dates that reach the set limit in dayLimits.

3.3.2 System Data Workflows

Data will move through the cloud architecture across two primary operational workflows:

Workflow 1: Placing a Custom Order (Customer Side)

1. Item Selection: The customer will choose cake tiers, sizes, chiffon flavors, and frosting options on the Custom Cake Builder page within the Customer Portal.
2. Instant Price Math: Built-in JavaScript functions will calculate the total price estimate instantly on the screen.
3. Design Reference Upload: The customer uploads custom cake reference images directly to Firebase realtime database convert into url through the order form.
4. Receipt Submission: The customer submits their payment receipt screenshot to C&N Bakery's official Facebook Messenger page for manual staff verification.
5. Data Saving: The web application sends order details, payment status, and Firebase Storage image links to the orders node in the database.
6. Capacity Update: The database will add +1 to the booking count under the dayFulfillment node for that chosen date to keep daily kitchen capacity accurate.

Workflow 2: Managing & Updating Orders (Admin Side)

1. Account Verification: The bakery owner or staff member will open the separate Admin Portal link and log in through Firebase Authentication to confirm staff access.

2. **Viewing Live Orders:** The admin dashboard will fetch active order text records and product images from the orders node in the Firebase Realtime Database and display them on screen in real time.
3. **Updating Order Progress:** A staff member will click to update an order status (moving an order from Pending to Confirmed, or Baking to Ready).
4. **Live Sync & Email Alert:** Firebase will save the new status instantly, update the progress bar on the customer's order tracking screen, and trigger an automated email update to the customer.

3.3.3 External Integrations & Messaging Layer

To keep the platform fast and running costs minimal, two external tools will support the core web architecture:

- **Facebook Messenger Channel:** Will serve as the manual verification portal where customers submit payment receipts and high-resolution design photos. This setup will keep heavy payment verification files off the main database while making use of C&N Bakery's existing social media setup.

- **Automated Email Notification System:** Will connect to backend database updates to send automated order status and progress alerts to customer email addresses whenever staff update an order stage. The team configured three specific email notification triggers corresponding to order confirmation, downpayment verification, and final delivery or store pickup.

3.4 Tools and Technologies Used

To build, host, test, and manage the web platform, the research team selected a practical stack of web technologies, development software, and cloud tools. The following sections describe each tool and detail exactly how it will be used to develop and support the C&N Bakery platform.

Tech Stack Summary

Technology Category	Selected Tool / Technology	Specific System Role in the Project
Front-End Layer	HTML5, CSS3, JavaScript(ES6)	Will build user screens, control mobile layouts, and run real-time price

		calculations right in the browser.
Cloud Backend	Google Firebase (Auth & Realtime Database)	Will handle secure account logins and sync lightweight text records live across screens.
Front-End Hosting	Vercel Deployment Platform	Will host the front-end web application live on a global network over secure HTTPS.
Cloud Free Tier	Firebase Spark Plan	Will provide zero-cost cloud database hosting within set free-tier usage limits.
Email Integration	EmailJS / Automated Email	Will send automated order confirmation, progress status emails and

		pickup/delivery to customer inboxes.
Development Workspace	Visual Studio Code (VS Code)	Serves as the primary code editor for writing, organizing, and formatting project files.
Version Control	GitHub	Tracks development progress, saves project milestones, and maintains cloud backups.
Testing & Monitoring	Firebase Console	Will be used to inspect live database entries and test security rules.

3.4.1 HyperText Markup Language (HTML5)

HTML5 is used to build the actual structure of every page in the system, both on the customer side and the admin side. Instead of relying on a framework, all pages are built using standard HTML5 elements. This includes:

- **Custom Cake Builder Forms** - input fields, dropdowns, and text boxes where customers type in cake details like size, flavor, inscriptions, and even paste a link if they have a design reference in mind.
- **Order Tracking Screens** - simple cards that lay out the order info and show the customer where their order currently stands.
- **Admin Control Panels** - tables and forms that let bakery staff manage the menu, set how many orders can be booked per day, and go through new orders as they come in.

3.4.2 Cascading Style Sheets (CSS3)

Cascading Style Sheets (CSS3) will control the visual styling, colors, fonts, and layout responsiveness across all screens. The development team will write custom CSS3 rules to achieve:

- **Mobile-Responsive Layouts:** Flexible grid and layout rules that ensure complex admin tables automatically adjust into simple, stacked card views when opened on smartphones.
- **Visual Status Indicators:** Clear color-coded badges applied to the 4-stage progress tracker to distinguish between Pending, Confirmed, Baking, and Ready for Pickup orders.
- **Interactive Screen Feedback:** Hover effects, button highlights, and instant color updates on pricing fields to make user actions clear and noticeable.

3.4.3 JavaScript (JS)

JavaScript is the main language handling all the logic and interactivity – no frameworks or libraries like React or jQuery, just plain JS working directly with the Firebase SDK. It's mainly responsible for three things:

- **Live Price Calculation Script:** Math scripts that will calculate price estimates in real time as customers select cake sizes, layer counts, flavors, and add-ons, updating the total price on screen instantly without reloading the page.
- **Date-Locking Calendar Engine:** Scripts that read active booking counts from the database, compare them against daily kitchen capacity limits, and automatically block out fully booked dates on the checkout calendar.

- **Real-Time Cloud Communication:** Background routines that will send user inputs to Firebase as structured text records and stream live order updates directly to the Admin Dashboard.

3.4.4 Firebase Authentication

Firebase Authentication will manage user account security for the platform, eliminating the need to write custom password encryption or account verification code from scratch. The research team will use this service to:

- **Manage Customer Accounts:** Handle customer sign-ups, email logins, password protection, and active user login sessions on the main Customer Portal.
- **Enforce User Permission Levels:** Verify staff access credentials whenever someone tries to log in to the dedicated Admin Portal, blocking unauthorized access automatically.

3.4.5 Firebase Realtime Database

Firebase Realtime Database will serve as the cloud storage vault for the platform, organizing system records into structured text trees that sync live across connected screens without requiring manual page refreshes. It will be used to:

- **Store Lightweight System Records:** Keep product catalogs, cake options, pricing rules, user profiles, active orders, kitchen schedules, and sales summaries neatly organized as JSON text records across root nodes like users, products, categories, customization-options, orders, dayLimits, dayFulfillment, carts, featured-cakes, settings, and security question.
- **Enable Live Status Updates:** Show new orders on the Admin Dashboard the instant a customer completes checkout, and update the customer's order tracking screen live whenever staff change an order status.

3.4.6 Vercel Deployment Platform

Vercel will serve as the cloud hosting platform for the front-end web application. The development team selected Vercel to:

- **Host Web Screens Live:** Deliver both the public Customer Portal and the dedicated Admin Portal live over secure HTTPS connections.
- **Automate Deployment:** Connect directly to the GitHub repository to automatically build and deploy new web code whenever updates are saved.

3.4.7 Firebase Spark Plan (Free Tier)

The Firebase Spark Plan will provide zero-cost cloud backend services within set free-tier usage caps. To ensure the system stays safely within this free limit:

- **Text-Only Storage Strategy:** The cloud database will store only lightweight text, numbers, dates, image URLs, and order records.
- **Messenger Proof Submission:** Customers will send payment receipt screenshots directly to C&N Bakery's official Facebook Page Messenger inbox, while custom cake reference photos will be uploaded to Firebase realtime database as image links. This setup keeps database text records lightweight, fast, and safely within free limits.

3.4.8 Automated Email Service (EmailJS)

An automated email service will handle customer communications directly triggered by backend database updates. The team will use this tool to:

- **Trigger Status Emails:** Send automatic confirmation, downpayment verification, and pickup alerts directly to customer email addresses whenever order status changes are saved by staff.

3.4.9 Visual Studio Code (VS Code)

Visual Studio Code serves as the primary workspace for writing, organizing, and managing system source files throughout development. The team uses VS Code to:

- **Write and Organize Files:** Draft and organize HTML5 markup, CSS3 stylesheets, and JavaScript files within a single clean workspace.
- **Maintain Code Quality:** Use built-in formatting tools and error checks to spot syntax mistakes, unclosed page tags, and coding errors early during development.

3.4.10 GitHub

GitHub provides online cloud hosting for the project source code. The team uses these tools to:

- **Save Project Milestones:** Create clear progress snapshots (commits) after completing major web features across Sprints 1, 2, and 3.
- **Protect Working Code:** Write and test new features on separate code branches before merging them into the main project code, making sure existing features continue working smoothly.

3.4.11 Firebase Console Dashboard

The Firebase Console Dashboard will provide a web-based management panel to monitor database records, manage active user accounts, and adjust security settings during development. The dashboard will be used to:

- **Configure Security Rules:** Set up and test database access rules to guarantee that read and write permissions match user access levels.
- **Monitor Storage Limits:** Keep track of real-time database connections and storage space to make sure the platform stays well within Spark Plan free limits.
- **Inspect Live Data:** Review stored entries across database nodes, verify that live synchronization works during system testing, and correct backend configuration settings.

3.5 Data Gathering Techniques

To collect all business rules, daily kitchen needs, and software requirements for the web platform, the research team used three qualitative data gathering methods: Interviews, Direct Observation, and Document Analysis. The primary goal was to study C&N Bakery's manual, paper-based operations, find daily workflow bottlenecks, understand how staff calculate cake prices, and turn their existing business rules into exact software features.

DATA GATHERING METHODOLOGY
1. STRUCTURED INTERVIEWS -> Captured business rules, production capacities, and operational pain points from owner & staff .
2. DIRECT OBSERVATION ->Tracked physical kitchen workflows, order states, and customer interactions during peak hours.
3. DOCUMENT ANALYSIS -> Examined handwritten logbooks, paper receipts, pricing sheets, and historical product catalogs

3.5.1 Interviews

Interviews involved asking direct questions to key bakery personnel. This allowed the research team to gather practical operational details, learn what the staff needed day-to-day, and plan exact software features based on input from the people running the business.

The team conducted structured consultation sessions with C&N Bakery's staff, including the business owner , front-counter staff , and head kitchen bakers . These discussions focused on two main operational areas:

- **Pricing Rules:** How custom cake prices were calculated based on cake size, base flavors, icing layers, and extra decorations.
- **Kitchen Production Limits:** The maximum number of custom cakes the kitchen staff could safely bake and decorate in a single day without lowering product quality.

Summary of Interview Insights to System Features

Key Stakeholder Insight	Observed Operational Issue	Planned Software Implementation

Kitchen Baking Limit	The kitchen could only bake 3 to 5 custom cakes per day without losing quality.	The backend database will track orders per date under dayFulfillment and automatically lock out full dates on the checkout calendar when limits in dayLimits are reached.
Disorganized Order Details	Custom cake notes got lost in personal SMS text threads during rush hours.	All cake inscriptions, design notes, and photo reference links will be saved under a single unique Order ID in the database.

Planned System Integration

- Scheduling Capacity Rules:** Daily baking limits provided by the owner were converted directly into backend database logic. The system will track active orders for every calendar date and automatically disable checkout for dates

that hit their capacity limit, preventing the kitchen from being overbooked.

- **Unified Order Customization Rules:** To prevent custom order details from getting lost in personal text messages or loose paper slips, the platform will collect all cake dimensions, custom text inscriptions, design notes, and photo reference links during checkout, saving everything under one unique Order ID.

3.5.2 Observation

Direct observation involved watching real daily work as it happened in the bakery. This allowed the research team to record actual task routines, work habits, and processing times without interrupting the staff during business hours.

The research team spent time inside C&N Bakery during busy morning and weekend hours. The team watched how counter staff greeted walk-in customers, wrote down paper orders, tracked baking progress in the kitchen, and calculated end-of-day sales totals.

Through direct observation, the team identified three main operational slowdowns:

- **Task Visibility Gaps:** Kitchen bakers relied on verbal reminders or loose paper notes, making it hard to keep track of which cakes were pending, currently baking, or ready for pickup.
- **Manual Calculation Delays:** Counter staff spent extra time searching for base prices on paper lists and calculating custom add-ons on handheld calculators while customers waited in line.
- **Manual Sales Summaries:** The owner calculated daily income by hand-totaling stacks of physical paper receipts at the end of each business day.

Planned System Integration

- **Order Tracker (4-Stage Pipeline):** Watching how kitchen staff tracked baking tasks directly guided the layout of the Admin Dashboard. The system will feature a 4-stage visual progress tracker (Pending, Confirmed, Baking, Ready), allowing staff to update and monitor order statuses in real time.
- **Automated Sales Reports:** Observing how the owner compiled paper bills to compute daily earnings guided the analytics features. The database will categorize completed sales automatically, generating daily, weekly, and monthly sales

summaries on the admin screen without manual receipt counting.

- **Categorized Menu Navigation:** Watching budget-conscious walk-in customers guided the online storefront layout. The website will include category filters and price-sorting options so customers can quickly browse items that match their budget.

3.5.3 Document Analysis

Document analysis involved reviewing physical business records, forms, logbooks, and paper files. Analyzing these real-world records helped the development team understand established business rules, pricing data, and record-keeping habits.

The research team collected and reviewed C&N Bakery's physical paper logbooks, handwritten order slips, printed price reference sheets, and product catalogs. Reviewing these records provided the exact math formulas, form fields, and business rules the bakery had used for years.

Document Analysis to Database Mapping

Bakery Paper Record	Data Extracted From Document	Planned System Equivalent

Handwritten Order Slips	Form fields for size, flavor, tiers, and custom text inscriptions.	Digital Custom Cake Builder forms and checkout input fields.
Printed Price Reference Sheets	Base prices per size and add-on costs for extra tiers and flavors.	Automated JavaScript price calculator running in the browser.
Highlighter-M arked Receipts	Colored markers showing paid deposits versus unpaid balances.	Automated database status tags (<i>Pending, Confirmed, Baking, Ready</i>).

Planned System Integration

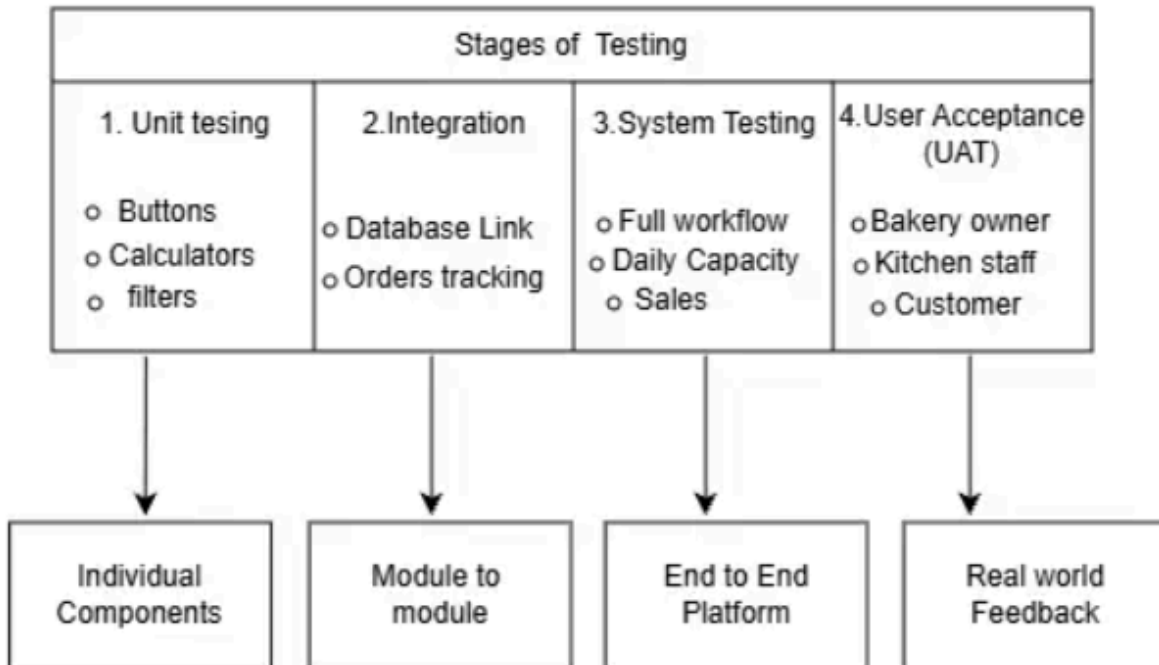
- **Digital Form Layout:** The physical structure of handwritten paper order forms—specifically fields for cake dimensions, tier counts, flavor options, and icing types—served as the model for designing the online Custom Cake Builder.

- **Automated Price Calculator:** The team converted handwritten price sheets into simple math rules inside the front-end JavaScript code. This will ensure that any combination of cake size, base flavor, and decorative options produces an instant price calculation that matches C&N Bakery's official rates.
- **Structured Data Organization:** Analyzing past product catalogs guided the design of the Firebase database structure, ensuring menu items will be organized logically into clear collections (such as products, categories, and customization-options).
- **Standardized Status Tags:** Reviewing old paper receipts showed that staff used colored highlighters to separate paid orders from partial down-payments. To eliminate confusion, the database will assign standardized status tags to every transaction, replacing physical highlighters with automated status fields.

3.6 Testing Procedures

To make sure the C&N Bakery web platform runs smoothly during daily business hours, processes customer orders safely, and handles real-time updates accurately, the research team will

execute a structured, multi-stage testing plan. Testing will cover every core system module—from front-end custom cake price calculators to real-time database synchronization between the customer storefront and the admin dashboard.



Overview of Testing Stages

- **Unit Testing:** Will test small individual parts on their own, such as buttons, price calculation scripts, menu category filters, and form text boxes.
- **Integration Testing:** Will check how different parts pass data back and forth, such as connecting customer checkout

choices to Facebook Messenger for payment proof and syncing calendar capacity limits in real time.

- **System Testing:** Will test the complete platform end-to-end, starting from guest menu browsing down to admin sales analytics recording.
- **User Acceptance Testing (UAT):** Will evaluate the finished website with the bakery owner, counter workers, kitchen staff, and real customers to confirm that the platform is easy to use and fits daily bakery operations.

3.6.1 Unit Testing

Definition

Unit testing will involve examining individual code components and user interface elements in isolation. The main goal will be to verify that every small feature works properly on its own before connecting it to other parts of the website.

How research team will conduct it:

The development team will test small interface functions across three main operational areas:

1. **Testing the Custom Cake Price Calculator:** The team will run math test scripts inside the cake builder by picking various layer options, tier sizes, chiffon base flavors (e.g., Vanilla, Chocolate, Mocha), frostings, and printable toppers. The team will compare the automatically calculated total price and the 50% downpayment requirement directly against C&N Bakery's official printed price list to confirm math accuracy.
2. **Testing Menu Budget Filters:** The team will click through each budget filter option (P0-P199, P200-P500, and P501+) and category tab (Classic Cakes, Cake Collections, Kakanin, Pastries) to confirm that the catalog displays only items matching those exact conditions.
3. **Testing Form Validation Rules:** The team will test input forms by intentionally leaving required fields blank (such as contact numbers), typing invalid email formats, or submitting incomplete checkout details. This will confirm that the system blocks improper submissions and displays simple, helpful error messages.

Planned Unit Test Cases

Tested Component	Test Input / Action	Planned Test Objective	Expected System
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			Behavior
Cake Price System	Selects 2 Tiers + Chiffon Base + Vanilla Frosting + Printable Topper.	Verify that JavaScript accurately calculates the total estimated price and the 50% required downpayment.	Calculated total price and 50% downpayment match the official bakery price.
Budget Filter	Click the ₱199 – ₱501+ budget filter button.	Verify that the storefront hides products outside this price range.	Displays only menu items priced between ₱199 and ₱501+.
Category Filter	Click the Kakanin menu category tab.	Verify that the product catalog filters items correctly by category.	Shows only Kakanin items and hides Classic Cakes and Pastries.

Form Validation	Submits checkout form with the phone number field left blank.	Verify that input validation blocks incomplete submissions.	Blocks submission, highlights the field in red, and prompts for a valid phone number.
Guest Access Control	Unauthenticated user clicks Add to Cart or Customize Cake.	Verify that protected features require account login.	Blocks action and opens the authentication modal (Log In / Sign Up).
Image Upload Tool	Uploads a custom cake reference photo in .jpg or .png format.	Verify that the file uploader accepts valid image attachments.	Accepts file, renders a preview thumbnail, and attaches image data to the order summary.

3.6.2 Integration Testing

Definition

Integration testing will focus on checking how different software modules pass data back and forth. This will ensure that data flows smoothly across web pages, user accounts, external communication channels, and backend cloud databases without errors or missing details.

How research team will conduct it:

The research team will test data communication across the front-end user interface, authentication services, and the Google Firebase Realtime Database:

- **Account Registration and Profile Syncing:** The team will test the link between customer sign-up forms and user database records. When a new user registers, the system will create their account profile, store contact details, and link future orders to their personal profile history.
- **Real-Time Calendar Capacity Updating:** The team will test the link between customer checkout choices and the calendar database. Tests will verify that placing an order automatically updates the active booking count for that selected date in real time.

- **Manual Payment & Messenger Verification Link:** The team will test the link between checkout payment instructions (GCash QR / Bank Details) and staff order management. When a customer sends a payment screenshot via Facebook Messenger, staff will review the proof and update the order status in the New Orders tab on the admin dashboard.
- **Instant Order Status Synchronization:** The team will test live synchronization between the Admin Dashboard and the customer's order tracker screen. When staff update an order stage tag (changing from Preparing -> Baking -> Packing -> Delivery), the update will reflect instantly on the customer's order stepper without needing a manual page refresh.

Integration Testing Scope

Integration Focus	Testing Procedure	Expected System Behavior
Auth & Profile Link	Register a new user account, log in, and check the stored database user node.	User profile details save securely and automatically link to future customer order records.

Checkout & Calendar Link	Place test orders for a specific date and monitor calendar capacity counters in the database.	Daily order counts increase live; once capacity is full, the date locks automatically on checkout.
Admin Verification Link	Staff review payment screenshot proof from FB inbox, then click Confirm in New Orders.	Order moves from New Orders to Active Orders, status changes from UNPAID/PENDING, and an automated email is sent to the customer.
Dashboard & Tracker Link	Change order status stage tags on the Admin Dashboard and observe the customer's profile stepper.	Customer's live order stepper updates instantly without requiring a browser refresh.
Cancellation Link	Customer clicks Cancel Order while	Order updates to cancel in the

	order status is still Pending.	database, remove item from active kitchen queue, and updates admin logs.
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3.6.3 System Testing

Definition

System testing will evaluate the entire application from start to finish as a complete platform. This will ensure that all business workflows—from initial customer product selection down to admin sales logging—operate smoothly without system failure.

How research team will conduct it:

The research team will execute complete operational test runs to evaluate overall system stability:

1. **Full Order Placement Workflow:** Team members will simulate customer ordering by picking items or designing a custom cake, picking an available delivery/pickup date, following instructions to send payment proof via Facebook Messenger, and checking if order details land accurately in the Admin Dashboard.

2. **Daily Booking Limit Enforcement:** To test calendar locking features, the team will place continuous test orders on a single date until reaching the daily capacity limit. The team will verify that the system locks that calendar date automatically for any new checkout attempts.
3. **Automated Sales Reporting Workflow:** Team members will progress test orders through all stages until marking them as Completed. The team will confirm that completed transaction values automatically update the admin sales summary dashboard (Monthly Revenue Trends, Product Category Split, and Top Selling Products).

End-to-End System Workflow Map

Phase	Customer Side Workflow	Bakery Admin / Staff Side Workflow	System & Database Action
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Step 1: Exploration & Auth	Browse homepage, catalog, and FAQs. System prompts to sign up/log in when attempting protected actions (Cart, Custom Cake, Checkout) .	Maintains current menu items, pricing, and category listings via Menu Management.	Validates authentication credentials; session state unlocks user portal tools.
Step 2: Selection & Customization	Adds catalog items to cart or configures custom cake options (layers, sizes,	Pre-configures active options and base prices for custom cakes in admin settings.	Calculates live price estimates dynamically based on selected attributes.

	flavors, frosting, toppings, reference photo upload) .		
Step 3: Scheduling & Checkout	Reviews cart items, fills in customer information, selects fulfillment method (Delivery/Pi ckup), date, time, and payment method (GCash/Bank Transfer/COD) .	Configures daily capacity limits, calendar dates, and pickup/deli very availabilit y toggles.	Checks database daily booking limit; blocks selection if fully booked. Displays GCash QR / Bank details.

Step 4: Payment Proof & Verification	Submits order; sends proof of payment screenshot manually to the bakery's official Facebook Messenger page.	Checks FB Messenger inbox for payment proof; reviews order in New Orders tab and chooses to Confirm or Cancel.	Generates new order entries marked as UNPAID / PENDING. Sends email notification to customer upon confirmation/cancel lation.
Step 5: Fulfillment & Tracking	Tracks real-time order status stepper (Preparing -> Baking -> Packing -> Delivery) via Profile	Updates stage status tags on Active Orders list as production progresses.	Realtime Database syncs status changes instantly to customer display without refresh; triggers progress emails.

	screen. Can cancel if not yet confirmed.		
Step 6: Completion & Analytics	Receives order; item moves to Order History where the customer can leave a rating and review.	Staff marks order as Completed on the Admin Dashboard.	Automatically records revenue in Sales Analytics (monthly, yearly, category trends). Updates public ratings.

3.6.4 User Acceptance Testing (UAT)

Definition

User Acceptance Testing (UAT) will be the final evaluation phase where real-world users—including the bakery owner, counter staff, kitchen bakers, and bakery customers—will evaluate the

platform to confirm usability, screen clarity, and operational fit.

How the research team will conduct it:

The research team will conduct structured testing sessions with representatives from all primary user groups using standardized evaluation tasks and feedback forms:

- **Customer Portal Usability Evaluation:** A selection of bakery customers will complete realistic tasks, such as browsing the menu catalog, applying budget filters, building a custom cake, checking price totals, submitting checkout details, and tracking order progress. The team will observe interactions to confirm that page navigation is simple and intuitive.
- **Administrative Operations Evaluation:** The bakery owner, counter staff, and kitchen bakers will interact with the Admin Dashboard. They will practice checking incoming orders, verifying payment receipts from FB Messenger, updating production status tags, adjusting daily capacity limits, and viewing sales graphs to ensure the interface supports daily store operations.

Planned UAT Participant Group

Evaluator Category	Primary Evaluation Tasks
Bakery Owner	Reviewing sales summaries, setting store rules, and managing product listings.
Counter Staff	Reviewing incoming orders, checking payment receipts, and updating status tags.
Kitchen Bakers	Monitoring active baking schedules and tracking daily production volume.
Bakery Customers	Customizing cakes, filtering product menus, and tracking order progress live.

3.7 Evaluation Criteria

To measure the success, overall stability, and operational value of the C&N Bakery web platform, the research team established an

evaluation framework based on five software quality standards:
Functionality, Usability, Reliability, Efficiency, and Security.

This framework outlines the testing rules and target performance metrics that will be used to evaluate both the cloud database and the front-end user experience for customers and bakery staff.

Evaluation Framework				
Fuctionality	Usability	Reliability	Efficiency	Security
Features	Navigation	Crash-proof	Fast speeds	Data safety
Calculations	Mobile view	Live sync	Instant logs	

5 System Evaluation Criteria

1. **Functionality:**Evaluates whether every feature, price calculator, and admin tool executes as intended without operational errors.
2. **Usability:** Assesses screen navigation clarity and ease of use across mobile devices, tablets, and desktop browsers for customers and store staff.
3. **Reliability:** Verifies that the site stays up, recovers from errors, and keeps user data safe during weak internet connections.

4. **Efficiency:** Measures system execution speeds, database synchronization times, and workflow automation compared to manual paper processes.
5. **Security:** Validates access control rules, user authentication, and data privacy safeguards to protect business records against unauthorized access.

3.7.1 Functionality

Functionality testing will confirm that every feature, total cost calculation, and admin button executes accurately according to our project design.

How the research team will test it

We will run test scenarios on both the main customer storefront and the admin dashboard to confirm every tool works cleanly under normal usage.

Key Focus Areas

- **Customer Storefront Features:** We will verify that catalog category filters work properly and that the Custom Cake Builder calculates costs accurately when users choose different sizes, cake bases, and designs.

- **Admin Dashboard Features:** We will check that the admin screen builds clear sales summaries and lets staff push orders through every kitchen stage without errors.
- **Order Tracking & Notifications:** We will confirm that order status updates sync live on the customer's screen and trigger automatic email updates.

Functionality Acceptance Goals

Feature Evaluated	Test Action	Expected Acceptance Criteria
Cake Price Calculator	Pick an 8-inch cake + 2 tiers + custom icing details.	Calculated total price will match the official bakery pricing sheet exactly.
Order Status Tracker	Staff updates an order tag from <i>Baking</i> to <i>Ready for Pickup</i> .	Status tag will update live on both administrative and

		customer tracking views.
Catalog Budget Filter	Click the ₱200 - ₱500 price filter button.	The page will show only products priced between ₱200 and ₱500.

3.7.2 Usability

Usability testing will check how easy and straightforward the site is for real people to use without needing manual or technical assistance.

How the research team will test it

We will track how smoothly customers and staff complete routine tasks across mobile phones, tablets, and desktop browsers.

Key Focus Areas

- **Customer Checkout Flow:** We will confirm customers can pick items, customize a cake, choose pickup/delivery times, and view payment details without confusion.

- **Bakery Staff Interface:** We will review admin dashboard readability during busy kitchen shifts to ensure buttons are large enough for staff to tap quickly on mobile screens with one hand.

3.7.3 Reliability

Reliability testing will show whether the platform stays stable, retains order details, and keeps working smoothly even when internet speeds drop.

How the research team will test it

We will run traffic stress tests and network disconnect tests to confirm the cloud database saves order records safely during outages or high website traffic.

Key Focus Areas

- **Handling Simultaneous Orders:** We will send multiple test orders at the exact same time to confirm the database handles concurrent requests without freezing or dropping data.

- **Saving Inputs During Disconnections:** We will check if customer form details stay saved in the browser if their internet connection drops midway through checkout.
- **Automatic Screen Reconnection:** We will confirm that the live order tracking page reconnects automatically once the user's internet drops and comes back, updating the screen without needing a manual refresh.

3.7.4 Efficiency

Efficiency measures how much time the platform saves on price calculations, record updates, and task tracking compared to manual paper logbooks.

How the research team will test it

We will measure system load times and feature speeds using browser developer tools, then compare those times directly against traditional paper order taking.

Efficiency Performance Benchmarks

Operational Task	Manual Paper Baseline	Target System Speed Metric	Verification Tool / Method
Calculating Cake Estimates	minutes (Manual lookup & calculator)	Instant second live calculation	Front-end JavaScript performance logging
Kitchen Task Tracking	Verbal calls or loose paper slips	Instant seconds cloud database sync	Firebase Console real-time monitoring
Checking Date Availability	Manual paper logbook checks	Instant real-time calendar date locking	Chrome DevTools Network panel timing

3.7.5 Security

Security testing will check access control rules and privacy safeguards to keep customer details, order records, and sales summaries safe from unauthorized viewers.

How the research team will test it

We will perform security checks to verify that accounts, business revenue records, and personal customer information remain secure.

Key Focus Areas

- **Admin Area Protection:** We will confirm that regular customer accounts cannot access or edit admin panel pages by typing secret URLs directly into the browser bar.
- **Form Input Safety:** We will test text fields to make sure bad scripts or code cannot be injected through public forms.
- **Business Data Safety:** We will verify that sales reports, revenue figures, and private customer information remain completely hidden from the public and accessible only to verified staff logins.

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CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The Custom Cake Ordering Platform is a web-based, mobile-friendly application created to simplify ordering custom cakes, manage daily baking schedules, and track sales performance for C&N Bakery. The system helps the bakery owner and staff organize custom cake details, set daily order limits, track baking progress in the kitchen, and view clear sales reports. By replacing paper order slips and unstructured chat messages with a centralized digital database, the app makes daily store operations much more organized.

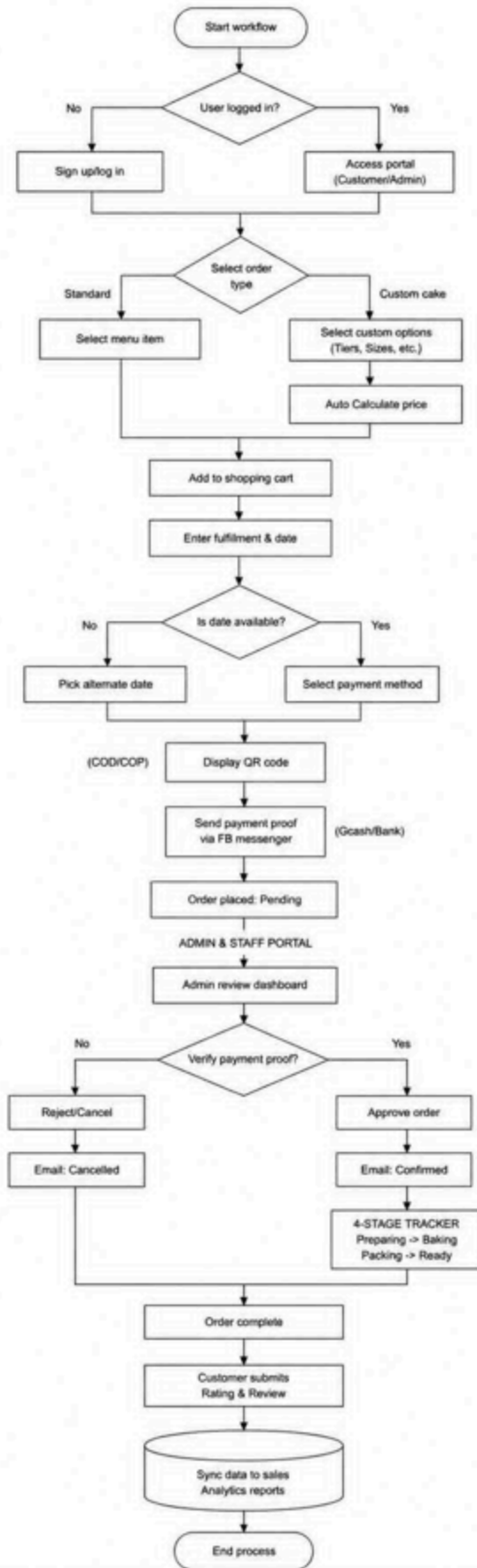
Unlike basic store registers that only record counter sales, this platform includes an automatic price calculator, a calendar booking control system, and visual sales reports. When customers pick their cake options—such as the number of tiers, sizes, flavors, and design details—the system calculates the order price right away. It also uses past sales data to build simple dashboards that show total earnings, popular cakes, and sales trends. Overall, the platform gives C&N Bakery a straightforward

tool that improves kitchen workflow, prevents double-booking orders, helps with ingredient planning, and securely saves customer records.

The major objectives of the developed system are as follows:

- Automatically record orders, calculate prices, and save custom cake details.
- Prevent kitchen overbooking by automatically locking fully booked calendar dates.
- Provide a simple 4-stage tracker for kitchen staff to manage baking progress.
- Generate clean visual charts and daily sales summaries.
- Point out best-selling menu items and popular cake designs.
- Help the bakery owner make practical business choices based on real sales figures.

Figure 4.1 illustrates the overall workflow of the developed system.



The process begins when a user accesses either the Customer Portal or the Admin Portal. Once logged in, customers can build their custom cakes, check if their desired delivery or pick-up date is available, and place their orders. Authorized staff members then review incoming orders and payment receipts, manage kitchen work using a four-stage progress tracker, adjust schedule limits as needed, and view sales reports generated from completed transactions. All results are presented through simple visual charts and summaries to help the business owner easily understand sales performance.

4.3 System Development Methodology

The team used the Agile software development method to build the C&N Bakery platform. Agile was chosen because it let the team develop the system in short, flexible cycles. This made it simple to adjust features based on feedback from the bakery owner and kitchen staff, making sure the final web application fit their daily store routine.

Using Agile, development was broken down into short phases called sprints. At the end of every sprint, a working section of the system was finished, tested, and reviewed before starting on the next set of features. This step-by-step process made it easy to catch bugs early and kept development on track.

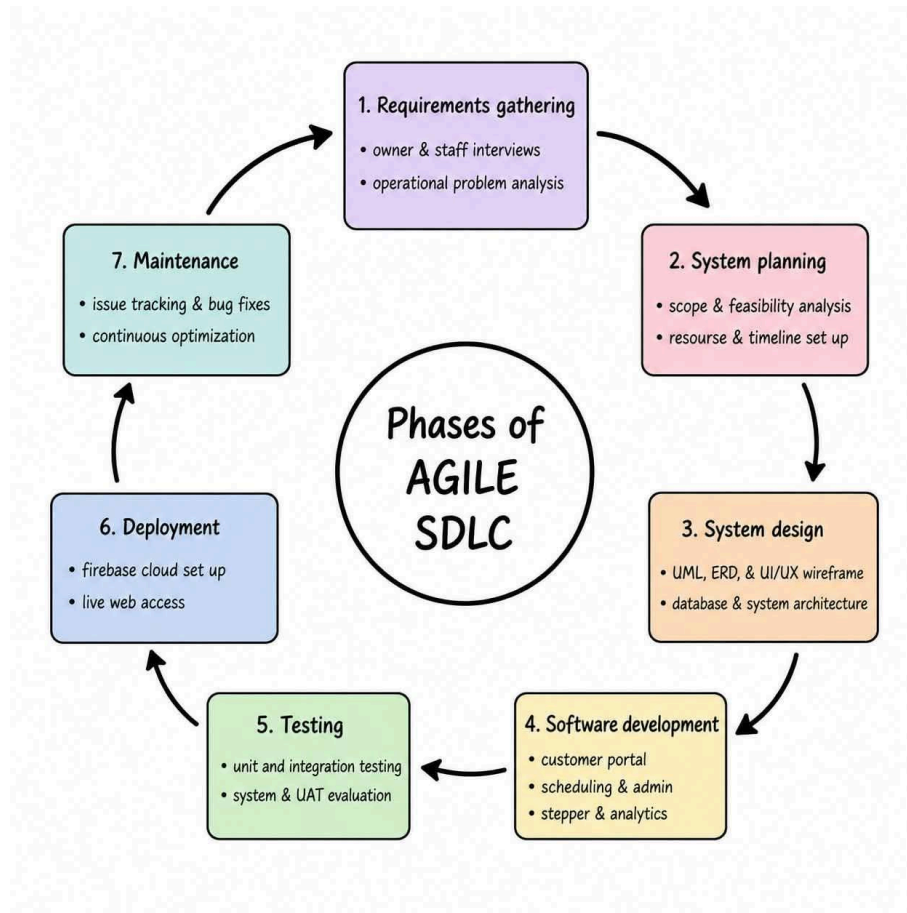


Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.

1. Requirements Gathering

Development started by talking directly with the bakery owner and kitchen staff through interviews and store observations. The team looked into daily operational struggles like lost paper order slips, mixed-up cake details, unorganized chat messages, and overbooked baking days. The feedback collected helped define the core features needed for the application.

2. System Planning

After identifying the bakery's needs, the team mapped out the project scope, feature list, project timeline, and development tools. This step made sure the planned platform was practical to complete within the target schedule and available technical setup.

3. System Design

During this phase, the blueprint for the entire web application was created. The team prepared database layouts, process flowcharts, and screen designs. System diagrams and visual wireframes were drawn up to guide the coding stage.

4. Software Development

The coding phase focused on building the website using HTML5, CSS3, and JavaScript for the user screens, combined with Google Firebase for backend cloud storage and user logins. System features were built across three main development sprints:

- **Sprint 1 (Customer Portal & Custom Builder):** Built the public menu catalog, account registration pages, the automated cake price calculator script, and the checkout form.

- Sprint 2 (Smart Scheduling & Admin Dashboard): Developed the calendar booking limits, the Facebook Messenger payment verification setup, and the central administrative order management panel.
- Sprint 3 (Kitchen Tracker & Sales Analytics): Built the 4-stage kitchen tracker (Preparing, Baking, Packing, Ready for Pickup/Delivery), automated status email updates, customer rating forms, and visual sales summary reports.

5. Testing

Each completed feature was tested individually before being linked to the full platform. Integration testing verified that the front-end user screens pass data back and forth accurately with the Firebase cloud database. Finally, User Acceptance Testing (UAT) was conducted with the bakery owner, staff, and real customers to confirm the platform works reliably in actual store conditions.

6. Deployment

After clearing all tests, the web application was hosted online for live access. Front-end code files were published through web hosting, while Firebase database security rules were set up to restrict admin screens strictly to authorized bakery staff. The site was also checked on smartphones, tablets, and laptops to

ensure touch-friendly controls and smooth layouts across all screen sizes.

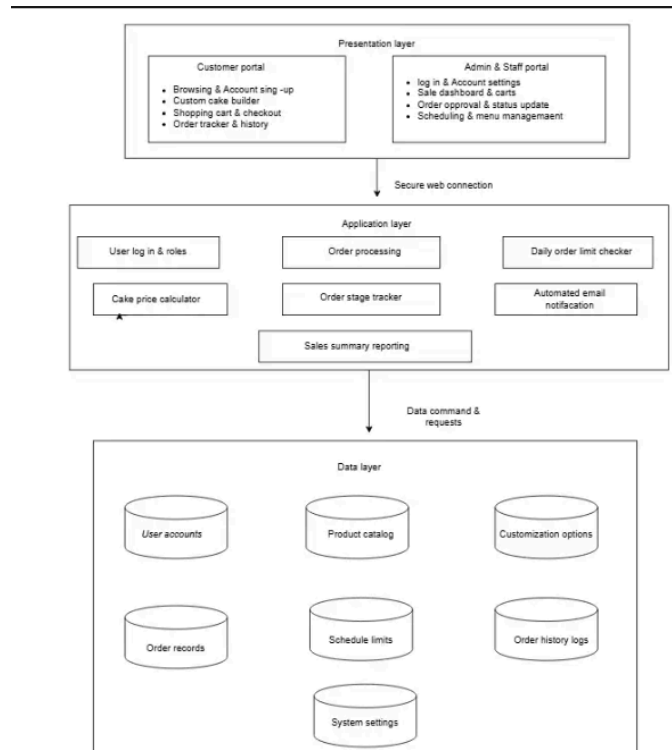
7. Maintenance

After launch, the platform is monitored to ensure stable online performance. Minor bugs reported during daily store use were fixed, database security rules were tuned, and small interface tweaks were made based on ongoing feedback from bakery staff.

4.4 System Architecture

The C&N Bakery platform runs on a three-tier setup. This design splits the system into three basic parts: the presentation layer (what users see), the application layer (the system logic), and the data layer (where information is stored). Structuring the system this way keeps the application organized, secure, and easy to maintain.

Figure 4.3 shows the overall architecture of the C&N Bakery ordering and management system.



The system architecture includes the following main parts:

Presentation Layer

The presentation layer includes all the screens that customers, bakery staff, and the owner interact with. It displays web pages, tracks active orders, and presents visual sales reports. Users access the application on smartphones, tablets, or computers using standard web browsers.

The main interface components are:

Customer Portal

- Browsing Pages: Home Page, Product Catalog, FAQ Page, Sign-Up Window, and Login Window.
- Custom Cake Builder: Interactive design screen with live price estimates.
- Shopping Cart & Checkout: Cart summary, pickup or delivery date and time selectors, daily order limit checks, and payment choices (Cash, GCash QR, and Bank Transfer).
- Customer Account Dashboard: Live order tracker, active orders list, order details modal, past order history with star ratings, and account settings.

Admin and Staff Portal

- Login Screen: Secure account access for staff members.
- Sales Overview Dashboard: Total revenue charts, sales grouped by category, top-selling products, and custom date range filters.
- Order Management Panel: Incoming orders queue for approval and a 4-stage kitchen order tracker.
- Schedule & Capacity Controls: Calendar view, daily order capacity editor, and pickup/delivery toggles.
- Menu Management Tools: Product editor, category editor, custom cake option settings, and featured cake controls.

- Order History Archive & Security Settings: Past order records and admin account recovery setting

Application Layer

The application layer contains the core rules and logic of the platform. It handles user logins, checks daily order limits, calculates custom cake prices, triggers automated emails, and compiles sales numbers for the owner.

Its main functions are:

- User Logins & Roles: Handles account sign-ups, password checks, and restricts admin pages to staff logins.
- Order & Cart Processing: Validates cart items, verifies fulfillment methods, applies the required 50% downpayment rule for custom cakes, and saves pending orders.
- Capacity & Calendar Validation: Checks requested pickup or delivery dates against daily limits and blocks dates that are already full.
- Cake Price Calculator: Computes total prices instantly as customers select cake tiers, sizes, chiffon flavors, frosting choices, and toppers.
- Order Stage Tracker: Updates kitchen stages (Preparing -> Baking -> Packing -> Ready for Pickup/Delivery) and handles order statuses (Pending -> Confirmed / Cancelled).

- **Email Notifications:** Sends automatic email status updates to customers whenever staff approve or update an order.
- **Sales Summary Service:** Combines past sales records to generate monthly, yearly, and custom-range performance reports.

The application layer acts as the middleman, processing requests from the website before reading or saving data in the cloud database.

Data Layer

The data layer stores all system records in a cloud database, keeping information organized and easy to fetch.

Stored records include:

- **User Accounts:** Customer profiles, admin/staff logins, security tokens, and contact details.
- **Product Catalog:** Standard cakes, kakanin, pastries, prices, descriptions, photos, and menu categories.
- **Customization Options:** Selectable sizes, flavors, frostings, toppers, and extra add-on costs for custom cakes.

- Order Records: Complete transaction details, special notes, delivery or pickup choices, scheduled dates, and payment types.
- Schedule Settings: Daily order capacity numbers and store pickup/delivery toggles.
- Order Logs & Reviews: Order status change history, customer star ratings, and written feedback.
- System Settings: Database security rules and general store settings.

4.5 Hardware and Software Requirements

To run smoothly, the C&N Bakery platform needs specific hardware and software setups. These specs ensure that the web application loads quickly, handles custom cake orders without lagging, updates calendar availability in real time, and generates sales reports without issues.

4.5.1 Hardware Requirements

Table 4.1 presents the minimum and recommended hardware specifications required to run and manage the platform. The recommended specs reflect the actual admin environment used by C&N Bakery staff , ensuring quick dashboard loading and responsive data processing.

Table 4.1: Hardware Requirements

Component	Minimum Specification	Recommended Specification (Admin Device)
Processor	Dual-Core 2.0 GHz Processor	12th Gen Intel® Core™ i5-12450H (2.00 GHz) or higher
Memory (RAM)	4 GB RAM	8 GB RAM
Storage	128 GB SSD	512 GB SSD (477 GB usable)
Display / Monitor	1366 \times 768 Resolution	Full HD (1920 \times 1080) Display

Graphics Card	Integrated Graphics	NVIDIA GeForce RTX 2050 (4 GB) / Intel® UHD Graphics
Internet Connection	10 Mbps stable connection	50 Mbps or higher fiber connection
Mobile Device (Client)	Android 10 / iOS 12 or higher	Android 13 / iOS 16 or higher

The recommended setup gives the system plenty of power to handle daily customer traffic, display interactive sales charts, and update kitchen tracking screens without delays.

4.5.2 Software Requirements

Table 4.2 lists the software tools and cloud services used to build, host, and run the application. The backend uses Google Firebase Realtime Database

(candn-bakery-default-rtdb.asia-southeast1.firebaseio.com)

, which safely stores user accounts, menu categories, custom cake options, daily order limits, and sales records.

Table 4.2: Software Requirements

Software / Tool	Type / Role	Operational Purpose
Windows 11 Home (64-bit)	Operating System	Used as the main operating system to build, manage, and test the website.
Visual Studio Code	Code Editor	Used as the primary code editor for writing HTML5, CSS3, and JavaScript, OOP files.
Google Firebase Realtime Database	Cloud Database Management System	Stores system data in cloud nodes including carts, categories, customization options, daily fulfillment, order limits, featured cakes, orders, products, reviews, security

		questions, settings, and users.
HTML5 / CSS3 / JavaScript/OP	Web Development Technologies	Used to build user screens, the automated cake price calculator, and admin dashboards.
Google Firebase Authentication	Security & Auth Service	Handles secure user sign-ups, customer logins, and administrative permission checks.
Web Browsers	Client Web Browser	Used by customers and staff to access the web application across desktop and mobile devices.
Git & GitHub	Version Control System	Used to manage source code updates, track code revisions, and maintain

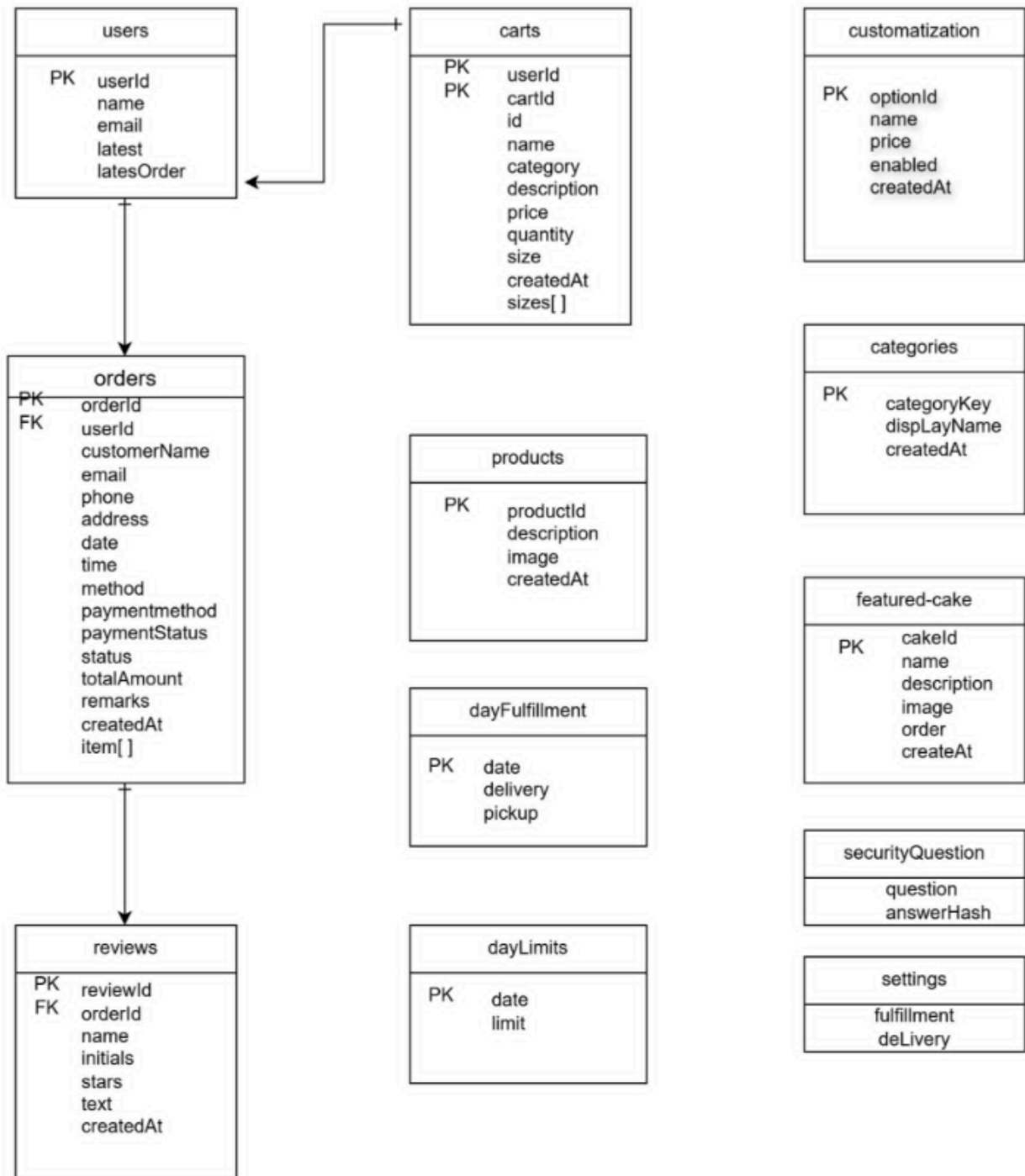
		system backups during development.
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4.6 Database Design

The database serves as the foundation of the C&N Bakery Web Application by organizing and storing customer accounts, menu items, custom cake configurations, active shopping carts, order workflow statuses, store settings, and customer feedback. A well-designed database structure ensures efficient data retrieval, prevents overbooking of custom orders, and maintains overall data integrity.

Rather than using a traditional relational table system (such as MySQL), the platform utilizes **Google Firebase Realtime Database**. In Firebase, data is structured as a hierarchical JSON data tree made up of collection nodes and sub-nodes. This setup allows updates to order statuses, shopping cart changes, and calendar availability to sync instantly between the client interface and the administrative dashboard.

FIGURE 4.4: FIREBASE DATA STRUCTURE DIAGRAM (ERD) OF C&N BAKERY



Primary Database Collection Nodes

The primary database collection nodes include:

- **carts Node:** Temporarily holds active shopping cart items, selected options, prices, and quantities per user.
- **categories Node:** Groups products into distinct menu sections (such as Classic Cakes, Customized Cakes, Kakanin, and Pastries).
- **customization-options Node:** Stores selectable attributes for custom cakes including designs, flavors, frostings, layers, and sizes along with their pricing.
- **dayFulfillment Node:** Controls daily pickup and delivery availability for specific calendar dates.
- **dayLimits Node:** Stores daily maximum order capacities set by management.
- **featured-cakes Node:** Manages showcase cakes rendered on the homepage gallery.
- **orders Node:** Contains detailed transaction logs, fulfillment schedules, payment statuses, ordered items, and 4-stage kitchen preparation states.
- **products Node:** Stores catalog items organized by category.
- **reviews Node:** Holds customer ratings, text feedback, and order reference keys.
- **securityQuestion Node:** Stores administrator password recovery questions and hashed answers.

- **settings Node:** Controls global store configurations such as delivery toggles.
- **users Node:** Stores registered customer accounts, user names, contact emails, and quick order history references.

Detailed Database Node Structures

1. carts Node

This node stores items added to a user's active shopping cart prior to checkout.

Field Name	Description
userId	User ID Key
cartId	Cart Item Key
id	Base Product Key
name	Product Name
category	Menu Category
description	Custom Notes
price	Unit Price
quantity	Item Quantity
size	Selected Size
createdAt	Added Date
sizes	Available Sizes

2. categories Node

This node organizes products into main menu sections on the store website.

Field Name	Description
categoryKey	Category Key
displayName	Category Name
createdAt	Created Date

3. customization-options Node

This node manages custom cake components for designs, flavors, frostings, layers, and sizes.

Field Name	Description
optionId	Option Key
name	Option Name
price	Additional Cost
enabled	Availability Status
createdAt	Create Date

4. dayFulfillment Node

This node manages daily fulfillment availability for delivery and pickup services.

Field Name	Description
date	Selected Date
delivery	Delivery Status
pickup	Pickup Status

5. dayLimits Node

This node limits daily custom order quantities to prevent kitchen overbooking.

Field Name	Description
date	Order Date
limit	Max Capacity

6. featured-cakes Node

This node manages featured cake listings shown on the landing page.

Field Name	Description
cakeId	Cake Key
name	Cake name
description	Product summary
image	Encoded image
order	Display Priority
createdAt	Created date

7. orders Node

This node records submitted customer transactions, payment methods, delivery details, and kitchen stages.

Field Name	Description
orderId	Order Key
userId	User ID Key
customerName	Customer Name
email	Email Address
phone	Contact Number
address	Delivery Address
date	Fulfillment Date
time	Selected Time
method	Fulfillment Type
paymentMethod	Payment Option
paymentStatus	Payment Status
status	Kitchen Stage
totalAmount	Total Price
remarks	Special Notes
createdAt	Submitted Date
items	Order Items

8. products Node

This node holds standard product catalog items grouped under category sub-nodes (cakes, customized_cakes, kakanin, pastries).

Field Name	Description
productId	Product Key
description	Product Details
image	Image URL
createdAt	Added Date

9. reviews Node

This node stores customer feedback and star ratings submitted after order completion.

Field Name	Description
reviewId	Review Key
orderId	Order Key
name	Reviewer Name
initials	User Initial
stars	Star Rating
text	Written Review
createdAt	Submitted Date

10. securityQuestion Node

This node stores backend administrative security parameters used for password recovery.

Field Name	Description
question	Security Question
answerHash	Encryption Hash

11. settings Node

This node controls system-wide operational toggles and global store parameters.

Field Name	Description
fulfillment	Service Settings
delivery	Delivery Toggle

12. users Node

This node stores registered customer profiles along with quick-reference links to their latest transactions.

Field Name	Description
userId	User Key
name	Full Name
email	Email Address
latest	Active Status
latestOrder	Recent Order

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary of findings, overall conclusion, and practical recommendations for the C&N Bakery platform. Throughout this project, the team focused on solving the main operational struggles of C&N Bakery—specifically managing custom cake details, preventing overbooking in the kitchen, tracking order progress, and viewing sales reports.

This chapter brings together the results from Chapter 4, reviews how well the project goals were met, and shares helpful recommendations for future developers and business owners. By combining online cake ordering, live tracking, automatic price calculations, calendar limits, and sales charts into one easy platform, this study shows how simple web tools can streamline daily operations for a local bakery.

5.2 Summary of Findings

Testing and implementing the C&N Bakery platform provided clear results that directly match the goals set at the beginning of the project:

- **Automatic Pricing and Faster Ordering:** The website successfully replaced messy paper order slips and disorganized social media chats. Prices for custom cakes are calculated automatically the moment a customer picks their cake size, tiers, frosting, and design add-ons. This cleared up price confusion, removed math errors, and made ordering way faster for customers.
- **Fewer Kitchen Mistakes on Cake Designs:** The custom cake builder stopped design mix-ups by letting customers submit exact details—like cake sizes, chiffon flavors, frostings, custom messages, and reference photo uploads—right on the site. Baking staff no longer have to guess or misread handwritten notes, which cut down on wasted ingredients and rejected cakes.
- **Easy-to-Find Store Rules and FAQs:** Adding an FAQ page with clear store policies, advance ordering rules, and payment steps gave customers quick answers to their questions. This saved front-desk staff from answering the same messages over and over again.
- **No More Overbooked Kitchen Days:** By setting daily order caps in the database, the site checks calendar availability before taking an order. During testing, trying to pick a fully booked date blocked the order and asked the customer

to pick a different day. This completely fixed the problem of taking on too many orders during busy seasons.

- **Clear Order Status for Kitchen and Customers:** The 4-stage kitchen tracker (Preparing -> Baking -> Packing -> Ready) gave bakers a clear task list and gave customers direct visibility into their order progress. Customers didn't need to call or text the store for updates anymore because updating a stage on the admin panel instantly updated the customer's screen.
- **Organized Pickup and Delivery Times:** Forcing customers to pick a specific date and time slot for pickup or delivery stopped scheduling mix-ups. Bakers now know exactly when a cake is needed, making sure orders are baked fresh right before pickup instead of sitting around or being forgotten.
- **Safer Customer Records and Cloud Backups:** Moving away from paper notebooks protects customer phone numbers, home addresses, and order histories from being lost, spoiled by kitchen spills, or seen by unauthorized people. Everything is stored safely in cloud database backups with secure logins for staff.
- **Instant Sales Numbers and Customer Reviews:** The visual sales dashboard turns completed orders into simple revenue charts, popular product lists, and date-filtered reports.

This saved the bakery owner from spending hours manually adding up paper receipts at the end of the month. Plus, the built-in rating feature gave the owner direct feedback on cake quality and service straight from customers.

- **Easy to Learn and Use:** Testing the site with bakers, store staff, and real customers proved the layout is clean and simple. Staff picked it up quickly without needing long training sessions, and user mistakes dropped significantly.

5.3 Conclusion

The custom ordering, tracking, and management platform built for C&N Bakery successfully achieved every single goal set for this project. It delivers a practical, easy-to-use digital tool created specifically around the day-to-day needs of a growing local bakery.

Before this system was built, C&N Bakery struggled with lost paper order slips, mixed-up cake design details, accidental double-booking during peak seasons, and time-consuming manual math for monthly sales. The new web application completely solved these operational headaches. By moving store management to a central cloud database, custom cake prices are calculated

automatically, daily kitchen limits are enforced strictly, and baking schedules run much more smoothly.

On top of that, live order tracking built real trust with customers and drastically cut down on constant status check messages. The admin panel simplified store operations by giving staff straightforward control over menu items, calendar booking caps, store settings, and instant sales reports.

In short, this project proves that accessible, well-designed web tools can transform small food businesses. The C&N Bakery platform successfully bridges the gap between everyday kitchen work and smart digital management—improving productivity, preventing costly mistakes, and giving the bakery owner clear numbers to make confident business decisions.

5.4 Recommendations

To build on what we've accomplished and keep the website useful for C&N Bakery as the business grows, we recommend the following system updates, research ideas, and daily store habits:

PROPOSED SYSTEM ROADMAP	
System Improvements	-> SMS Alerts and Direct payment gateway
Future Research	-> Machine Learning Demand Forecasting, Native App
Operational Strategy	-> Staff Training, Regular Backups, Security Audits

System Improvements

- **Direct Payment Gateway Integration:** Future developers should consider integrating automated payment gateways (such as PayMaya) directly into the checkout pipeline. This would enable instant payment verification without requiring staff to manually check payment screenshots sent via Facebook Messenger.
- **SMS Text Alerts:** While the current site sends email updates, adding direct SMS text messages would alert customers the second their cake is ready for pickup or out for delivery.

Future Research Directions

- **Predictive Sales Forecasting:** Future researchers can explore using machine learning models to look at past sales trends. This would help predict busy holiday rushes and suggest exact ingredient amounts to buy in advance.
- **Long-Term Revenue Study:** Running a 6-to-12-month study would help measure the system's long-term effect on store profits, staff working hours, and repeat customer rates.
- **Dedicated Mobile App:** Building a standalone mobile app for Android and iOS could give customers offline order drafts and better push notifications on their phones.

Implementation Strategies

- **Comprehensive Staff Training:** C&N Bakery management should run regular training sessions for all kitchen and front-desk staff. Consistent data entry habits such as updating order stages immediately as work happens in the kitchen are essential to keep customer tracking accurate.
- **Routine Database Backups and Security Audits:** System administrators should schedule regular database backups and periodically update security rules within Firebase to safeguard customer data and store records.
- **Regular Customer Feedback Reviews:** Business managers are encouraged to check the integrated reviews node regularly

to spot quality trends, address customer concerns quickly, and keep product offerings aligned with customer preferences.

5.5 Impact of the Study

This study provides real, practical value for both local business operations and the field of Information Technology. First, it offers a clear, working example of how small bakeries can digitize their ordering and kitchen tracking without spending money on expensive or overly complicated software. It demonstrates that local food enterprises can modernize their workflow using accessible, budget-friendly web tools.

Additionally, the project serves as a practical blueprint for future developers. It shows how to design a simple cloud database setup that smoothly handles custom product options, daily calendar booking caps, and live status updates without system lag.

The platform also creates a much better overall experience for customers. It makes custom cake ordering simple and transparent, gives buyers direct visibility into their order's progress, and removes the frustration of last-minute order cancellations caused by kitchen overbooking.

Finally, the system takes the guesswork out of daily store management by turning everyday transaction records into clear visual charts. This gives the bakery owner the exact numbers needed to restock inventory efficiently, highlight top-selling products, and run smart, data-backed store promotions.

5.6 Chapter Summary

Chapter 5 wrapped up the study by summarizing the main findings, reviewing what the project accomplished, and offering practical next steps for the C&N Bakery platform. The results proved that automating order intake, enforcing daily kitchen limits, providing step-by-step order tracking, and displaying visual sales charts significantly improved daily store operations and solved the bakery's biggest headaches.

By directly fixing the real-world operational struggles of C&N Bakery, this research provides a reliable foundation for future developers to build on, while showing how local food businesses can easily use modern web tools to stay organized and grow over time.